MirrorCut: Anytime-Valid Component Screening Under Pinned Mid-Run Commitments

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Abstract

An agent harness is a set of components - context policy, tool set, retry rule, guardrail - and teams decide which of them stay while evidence is still accruing. This paper asks what a single e-process can guarantee when those decisions are committed mid-run. Under independent level randomisation among the undecided components, bounded outcomes, predictable bets, a fixed task distribution and a frozen verifier, and no carryover within a pair or interference across rows, one e-process carried across non-revoking commitments controls the all-stage conditional null that the conditional mean increment is non-positive, at any stopping time, with family-wise error at most alpha over admissions; each component carries a second e-process for pruning with its own budget, so acting on both costs the sum. Within the class of corrections that agree with the screen up to a commitment time and never revoke an admission, a component whose conditional effect turns non-positive at that time keeps every admission banked before it, while the probability of any further admission after it is bounded by the evidence banked there; and measure, in one experiment of 10,000 replications where three procedures share the random numbers, that the two non-revoking repairs a deployer would try reduce that component's admission only from 11.71% to 9.84% and 11.20%. Run forward on a public agent benchmark, in a lane that drops no assigned attempt and stores the executed configuration and its digest for every row, the screen admits the prespecified expected-positive step-budget control at row 48 and leaves a cosmetic control undecided. Every ledger, pre-registration file and analysis script named in the paper is attached to this submission as supplementary material, with a manifest of SHA-256 digests.

1 The screening problem

An agent that resolves a software issue (Jimenez et al., 2023) or drives a terminal is not a model. It is a model inside a harness that decides what the model sees, which tools it may call, when a failed step is retried, and when the episode ends. The size of that layer is no longer in doubt: a recent runtime raises a fixed model from 83.1 to 92.1 on Terminal-Bench 2.1 by changing the execution system alone, and to the 95.3 of its title with its largest harness configuration (Qin et al., 2026); a dedicated benchmark now measures harness effects across models (Yao et al., 2026). Once a team accepts that the harness matters, it acquires a portfolio of switches, and with it the question this paper is about: which of them are actually paying, and which are inherited defaults that no measurement supports. The stakes are not hypothetical: replayed on public tau2-bench rows, the analyst who checks the interim numbers and stops at the first crossing ships a component that every error-controlled reading of the same ledger refuses (Table 21).

The question is a screening problem, and the field currently answers it in one of two ways. The careful way is to enumerate the full factorial and average: the closest prior study runs the complete 2^5 over five scaffolding components with up to ten seeds per cell, and reports that stacking components can hurt (Liu, 2026). The common way is to run a partial sweep and stop when the numbers look decided, a temptation that grows with the price of a rollout; a replay study of public benchmarks finds that the fraction of tasks needed before a partial run supports the completed decision ranges from 15% to more than 90% depending on the benchmark (Huang, 2026). Neither way carries an error guarantee for the decision it is used to make.

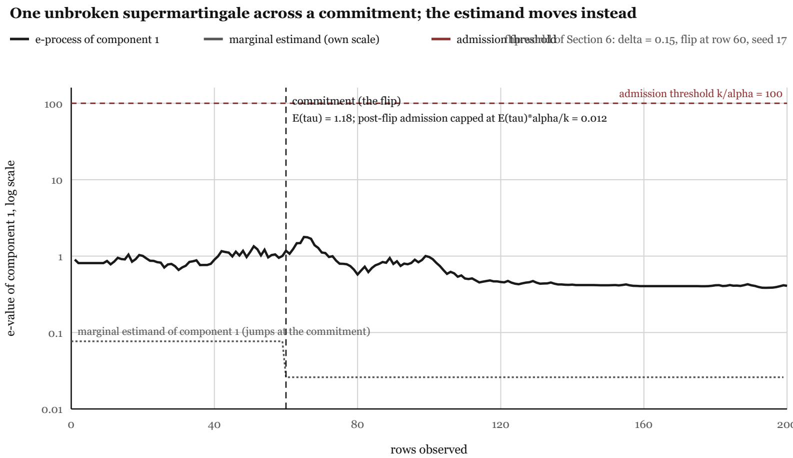

Figure 1. The paper's claim in one run, computed in the flip world of Section 6 (delta = 0.15, flip at row 60): the evidence process of the flipping component is carried across the commitment without restart and without correction, while the component's marginal estimand jumps sign at the commitment. In this world the process is not a supermartingale before the flip, since the component genuinely helps there; conditional on the evidence at the commitment, the continuation E(n)/E(tau) is a non-negative supermartingale after it. What the flip can still buy after the commitment is capped by the evidence banked at it, E(tau) times alpha over k; the exact figures for this run are printed on the canvas.

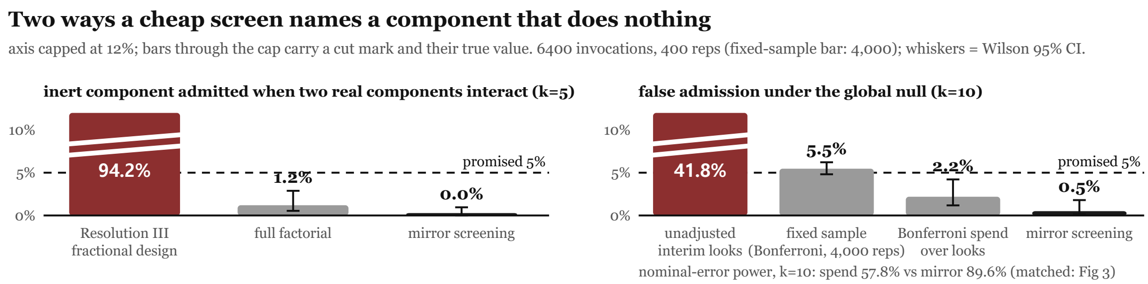

Figure 2. Two measured failure modes of the comparators, on the illustrative simulator specified in Section 5. Left: with two components genuinely interacting (synergy scenario, heterogeneous tasks, 6400 invocations), a Resolution III fractional design admits the inert component that carries their alias in 94.2% of replications. Right: under the global null at ten components, unadjusted interim looks raise the family-wise error rate to 41.8%. The randomised screen stays below the nominal level in both settings at the same budget.

Two properties of agent evaluation make this worse than it looks. First, outcomes are not stable: replaying identical inputs through the deployment used in this paper flips 11.7% of row-level verdicts, and consistency under semantically preserving perturbation is now itself an object of study (Raj et al., 2026). Second, the components interact, so an estimator that confounds a main effect with an interaction will not merely be imprecise, it will name the wrong component. We show below that this matters for the design a cost-pressed team would reach for: a Resolution III fractional design, the textbook answer to enumeration cost in industrial screening (Box, Hunter and Hunter, 2005) though not yet documented in harness studies, attributes an interaction to an inert component in most cases. A fold-over repairs that aliasing at equal total budget and Section 6 measures the repair; the design is examined not as the strongest baseline but as the shortcut actually reached for, and the failures that survive the repair, peeking and multiplicity with no pre-planned sample size, are the ones this paper is about. Measured on the illustrative simulator specified to the deployment above, the three incumbent practices fail at these rates: the fractional design admits the aliased inert component in 94.2% of replications, unadjusted interim looks inflate family-wise error to 41.8% at ten components against a nominal 5%, and the point-comparison acceptance rule of self-improvement loops false-adopts with probability 0.999 per round, computed exactly.

This paper proposes a procedure with three properties that the comparators evaluated here do not have together. It does not enumerate configurations, so its per-row execution cost is independent of the number of components; the invocations needed for a target power still grow with the number of components through the alpha/k threshold. It cannot confound a two-factor interaction with a main effect, by construction rather than by design resolution. And it is valid under continuous monitoring, so a team may look at the evidence whenever it likes and stop whenever it likes. The price is quantified rather than hidden: a few points of power at equal budget against a fixed-sample full factorial, decomposed exactly in Section 6.

The cost is stated as plainly as the guarantee. The per-row cost is O(1) in the number of components, but rows-to-decision scales inversely with the squared effect: at the stated simulator constants a component moving the failure mass by 0.15 needs about 2,200 invocations to admit, and one moving it by 0.05 about 19,857 (Table 9). When the component count is small and the effects are small, a fixed-sample full factorial is simply cheaper; what the screen buys is validity under continuous monitoring and a cost that does not grow with k, not free power. Both public runs in Section 8 land exactly where that table says they will.

Concretely, the paper contributes: (i) the conceptual result the title states - once a screen commits decisions mid-run, any procedure that agrees with the screen up to the commitment and does not revoke admissions inherits every admission the screen already made, and what remains bounded is only the additional probability of admission after the commitment (the class is stated precisely in Section 4), and the null that carries a single-supermartingale anytime-valid guarantee is the conditional one over reachable committed contexts; we prove control at any stopping time under that null, prove that a sign-flipping component's post-commitment admissions are confined to the evidence banked before its flip, and measure that per-epoch restarts and pinned re-screening leave the flip price standing (Section 4, Propositions 5 and following; Section 6); (ii) a three-line screening procedure whose cost per row is two invocations regardless of the number of components, with a composition result showing a self-improvement loop inherits the judge's guarantee whatever its proposer does, and a shipping protocol whose α-split confirmation covers the deployed configuration even where the conditional null does not (Sections 3-4); (iii) the shipping rule a practitioner actually needs: unpaired one-invocation screening by default on binary verifiers, mirror pairing exactly when an per-invocation log-growth criterion computed under the assumed increment law, computable from a pilot, says the variance saving pays - a criterion that predicts, cell by cell, when pairing beats unpaired screening, whose one measured failure under deliberate misspecification costs power, never validity (Table 8 prints the miss), and yields a budget-planning table (Section 6); and (iv) a measured account of how the two standard alternatives fail, together with a live deployment run in which the procedure pruned one guardrail at its harm process's α-level (Section 7), and public-benchmark runs of both the screen and the full shipping protocol with a task split reserved in advance (Section 8).

2 Related work and what is left

Harness effects. Yao et al., 2026 measures them across models; Chen et al., 2026 isolates scaffold components through controlled composition; Liu, 2026 reports destructive interaction between components under a full factorial; Qin et al., 2026 demonstrates how large the effect of a runtime can be. All four establish that component attribution is worth doing. None supplies a decision rule with an error guarantee, which is what a team needs before it removes a component from a production harness.

Optimising the harness versus certifying it. A fast-moving line of work searches for a better harness directly: Zhang et al., 2026 evolves it component-wise and names cross-component interference as the obstacle, Chen et al., 2026 co-evolves the harness with the reasoner, and Luo et al., 2026 lets a reinforcement-learning agent edit harness components. These proposers raise performance but attach no error control to what they keep, and one of them reports overfitting to its own evolution tasks. This is the gap we fill: a proposer of any kind supplies candidates, and our procedure is the certified gate that decides which of them may ship without adopting an inert or harmful component beyond a stated rate. The interference these methods fight is the same aliasing our per-row mirror difference removes by construction.

Anytime-valid inference for language models. The tool we use is shared with a recent line of sequential work, and the distinction is the problem, not the machinery. Li et al., 2026 stops a two-agent comparison as soon as the evidence suffices, but ranks a pair rather than screening many components under one error budget;

Ota et al., 2026 certifies an answer as the mode of a self-consistency distribution with an intersection-union e-process, the same construction we carry, but its object is an answer, not a component's effect; Hsu et al., 2026 builds a confidence sequence for a single model's capability on a fixed question set, not an attribution across interacting components. Our contribution is to pose harness-component selection as a family-wise, anytime-valid screening problem and to pair each run with its mirror as a variance device inside that screen.

Evaluation methodology. Kapoor et al., 2024 makes cost a first-class axis and Kapoor et al., 2025 supplies the infrastructure to compare agents under a cost budget. Miller, 2024 gives fixed-sample confidence intervals for evaluations, and Zhu et al., 2025 catalogues the ways agentic benchmarks are built wrongly. Our procedure is complementary: it inherits the reporting standards of that literature and adds a stopping rule that survives the analyst looking early.

What we do not claim as new. The mirror difference is an old device. Antithetic variates pair a draw with its reflection to reduce variance (Hammersley and Morton, 1956); simultaneous perturbation stochastic approximation perturbs every coordinate at once with Rademacher signs and takes a two-sided difference (Spall, 1992); mirrored sampling does the same inside evolution strategies; and folding a Resolution III design to reach Resolution IV is textbook (Box, Hunter and Hunter, 2005). Our contribution is not the estimator. It is the object of the guarantee: the identification of the conditional null as what a committing screen can promise, the single supermartingale that survives commitments and optional stopping, the confinement bound on what a sign flip can cost, and the measurement of how badly the standard alternatives fail on this problem.

Anytime-valid inference. The sequential tools are likewise not ours: power-one tests (Robbins, 1970), testing by betting (Shafer, 2019), safe tests (Grünwald et al., 2019), time-uniform confidence sequences (Howard et al., 2018), betting-based mean estimation whose plug-in bet family we adopt (Waudby-Smith et al., 2020), always-valid inference for A/B peeking, the industrial ancestor of our second headline failure (Johari et al., 2015), and the survey that fixes our terminology (Ramdas et al., 2022). The false-discovery-rate extension named in Section 9 is e-BH (Wang et al., 2020). Reading main effects off randomly drawn factor levels also has prior art and a published critique: random balance experimentation (Satterthwaite, 1959), whose defect was precisely the absence of an error-controlled decision rule on top of the random design; the e-process is that missing rule. Best-arm identification (Jamieson et al., 2013; Kaufmann et al., 2014) answers a different question, selecting the best of k alternatives rather than testing each of k components against its own null, and does not handle the paired factorial structure here. Adaptive data analysis (Dwork et al., 2014) controls the bias of reusing one holdout across adaptively chosen queries; our problem is the transpose - the queries are the fixed component set, and what the procedure's own commitments change is the world the estimand lives in. What none of this supplies is the application: a screening rule in which the randomisation, the pairing and the conditional null are matched to how agent teams commit component decisions mid-run.

One anticipation is worth making explicit. To a sequential statistician the conditional null is not exotic: a test supermartingale's guarantee has always attached to conditional means given the filtration. We do not claim to have invented the object. The claim is about the gap between that object and the question screening practice actually asks - would this component help my deployment, the marginal question - and the two coincide only until the first commitment. This paper measures the gap where it opens (a flipping component is admitted in 10.8% of runs, Section 6), proves which part of it no post-commitment correction can close (Section 4), and states the guarantee that survives, together with the reserved-split confirmation that answers the marginal question with fresh data.

3 Mirror screening

Write a harness configuration as a vector x in $\{-1,+1\}^k$, one coordinate per component, where +1 means the component is switched on. Let y(x, t) be the verified outcome of running the agent on task t under configuration x, and let the response decompose as a polynomial in x with arbitrary coefficients:

$$y(x, t) = \mu(t) + \sum_i \beta_i x_i + \sum_{i<j} \gamma_{ij} x_i x_j + \sum_{i<j<l} \tau_{ijl} x_i x_j x_l + \ldots + \varepsilon$$

Algorithm 1 Mirror screening

Input: components 1..k, task pool T, admission level α_A, pruning level α_P, confirmation level α_C with $\alpha_A + \alpha_P + \alpha_C \leq \alpha_{total}$; admit at k/α_A, prune at k/α_P; invocation budget B.

Before any screening row: choose the paired or unpaired arm from pilot data that is excluded from every reported e-process, and fix it. The arm is never chosen from the screening rows themselves.

State: for each component, two e-values initialised at one, and a decision flag.

Repeat until the budget is spent or every component is decided:

draw a task t from T and levels x uniformly from $\{-1,+1\}^k$; overwrite the coordinates of decided components with their committed level,

paired arm: run the agent on t at x and at the mirror of x, where the mirror flips only the undecided coordinates. Unpaired arm: run the agent once, on t at x,

form the per-component signal: paired $z_i = x_i\ (y(x,t) - y(\text{mirror},t))\ /\ 2$; unpaired $z_i = x_i\ (y(x,t) - c_n)$ with c_n the running mean outcome of earlier rows; clip z_i to [-1, 1] - a no-op in this setting, since the paired signal lies in [-1/2, 1/2] and the unpaired signal in [-1, 1] by construction, so the clip never binds and cannot bias the increment,

update only the components undecided at the start of the row, with the predictable bets λ^+, $\lambda^- = 0.1$ for the first two rows and $\text{clip}(\mu/(\sigma^2 + \mu^2 + 0.1), 0, 0.45)$ thereafter, μ the running mean of $\pm z_i$ and σ^2 its running variance:

$E^+_i \leftarrow E^+_i\ (1 + \lambda^+_i\ z_i)$; admit i when $E^+_i \geq k/\alpha_A$, committing $x_i = +1$,

$E^-_i \leftarrow E^-_i\ (1 - \lambda^-_i\ z_i)$; prune i when $E^-_i \geq k/\alpha_P$, committing $x_i = -1$,

retire i as undecided-inactive when $E^+_i \leq f = 0.05$: no error statement is made about it, and it is executed at the incumbent level -1 from the next row on,

if both directional thresholds cross on the same row, the admission takes precedence and the double crossing is recorded in the ledger,

apply every decision of a row simultaneously from the next row on; a decided component keeps its committed level in both executions of every later row, so it contributes zero to every later signal.

Output: the admitted set, the pruned set, the undecided-inactive set, and the evidence path of each component.

Finally, confirm: run a fixed-sample paired check of the assembled final configuration against the pre-screen baseline on fresh tasks. The screen's guarantee is per component under the conditional null; the confirmation is the statement about the configuration actually deployed, and is a mandatory step of the protocol, not an optional audit.

A note on the name before the mechanics: MirrorCut names the family after its distinguishing variance option. Two arms share one guarantee: the mirror-paired arm developed here, and the unpaired one-invocation arm specified and proved in Section 4; the chooser of Section 6 decides between them on variance, not validity. The procedure is three lines. For each row: draw a task t and draw the levels x independently and uniformly from {-1,+1}; execute the agent twice on the same task, once at x and once at the mirror -x; form the paired difference $d = (y(x, t) - y(-x, t))\ /\ 2$ and the per-component signal $g_i = x_i\ d$. Components already decided are held at their committed level in both executions, so they contribute nothing to the mirror difference, while still shaping the context in which the undecided components are read, which is what the conditional estimand tracks. Each component carries two test supermartingales, one betting that the component helps and one on the negation, updated with a bet size computed only from earlier rows:

$$d = (y(x, t) - y(-x, t))\ /\ 2, \quad g_i = x_i\ d$$

$$E_i(n) = E_i(n\text{-}1) \cdot (1 + \lambda_i(n)\ g_i(n)), \quad \lambda_i(n) \text{ predictable, } 0 \leq \lambda_i(n)|g_i(n)| < 1; \quad \text{admit i when } E_i(n) \geq k/\alpha$$

Nothing else is required. There is no design matrix to build, no resolution to check, no sample size to commit to in advance, and no enumeration of the 2^k configurations. The cost is two model invocations per row regardless of k, and the analyst may inspect E_i at any moment and stop at any moment.

An earlier construction paired each run with a fixed all-off champion instead of the mirror: unbiased, and it too cancels the task effect, but half of its budget re-runs an uninformative baseline and its signal carries the other components' effects at full variance where the mirror halves them (neither cancels them; those terms are odd). Section 6 prices this: the champion-paired arm recovers a fraction of what mirror pairing recovers at equal budget.

4 Why the procedure is safe, and exactly what is guaranteed

Proposition 1 (annihilation of even-order terms in the mean response)**.** *Write $m(x, t) = E[Y(x, t) \mid F_{n\text{-}1}, t]$ for the conditional mean response and expand it on the hypercube. Then the mean contrast $(m(x, t) - m(-x, t)) / 2$ equals $\sum_i \beta_i x_i + \sum_{i<j<l} \tau_{ijl} x_i x_j x_l + \ldots$: every even-order term drops out of the mean contrast, including the intercept, the task effect, and all two-factor interactions. The observed difference $d = (Y(x, t) - Y(-x, t)) / 2$ is that mean contrast plus centred execution noise, since the two configurations are separate stochastic invocations; the noise has conditional mean zero and therefore does not disturb the e-process null, but it is not cancelled by the mirror.*

Proof. Substituting -x into a monomial of order m multiplies it by $(-1)^{m}$; even monomials cancel in the difference, odd ones survive. The task effect is even of order zero, which is why the two executions must share a task. ☐

Proposition 2 (unbiased signal under full mirroring)**.** *While no component has been decided, the levels are iid Rademacher and every coordinate is mirrored, so $E[g_i] = \beta_i$ exactly: each surviving cross term contains at least two distinct independent signs and has mean zero.*

Proposition 3 (what the e-process guarantees, stated for the algorithm actually run)**.** *The betting strategy is the plug-in rule $\lambda = \mathrm{clip}(\hat{\mu}/(\hat{\sigma}^2 + \hat{\mu}^2 + c), 0, 0.45)$ with running estimates from earlier rows only, a stabilised empirical-mean bet in the GRAPA family of Waudby-Smith et al., 2020, with stabiliser $c = 0.1$ and $\lambda = 0.1$ for the first two rows before running estimates exist. It is computed from earlier rows only, and is clipped to [0, 0.45] while $|g_i| \leq 1$, so $1 + \lambda_i g_i > 0$ and each E_i is a non-negative process with $E[E_i(n) \mid \text{past}] \leq E_i(n\text{-}1)$ under its null. Ville's inequality then bounds the chance that E_i ever reaches k/α by α/k, and a union bound over components gives family-wise error at most α at any stopping time. Two separate families run at once: the admission processes control admitting a component whose effect is not positive, and the harm processes control pruning a component whose effect is not negative, each at level α; a reader who acts on both admissions and prunings is carrying 2α in total, and every table reports the two families separately against their own α. The futility floor ($E_i \leq 0.05$) is a budget device, not part of either guarantee: a component stopped for futility is reported as undecided-inactive, and no error statement is attached to that outcome.*

Proposition 4 (partial mirroring identifies the committed-context contrast)**.** *Let C be the set of components committed by row n at levels c, let U be the undecided set, and let the levels of U be drawn independently and uniformly while the mirror flips only the coordinates in U. Write $m(c, x_U, t) = E[Y \mid c, x_U, t, F_{n\text{-}1}]$ for the conditional mean response. Then for every i in U, $E[g_i(n) \mid F_{n\text{-}1}] = E_{t,\ x\text{<sub>}U} [x_i\ m(c, x_U, t)]$, which is one half of the randomised contrast between $x_i = +1$ and $x_i = -1$ in the committed context c, averaged over the task law and over the levels of the other undecided components. Interactions between i and a pinned component are folded into that coefficient; interactions between i and another undecided component average out.*

Proof. The pair shares the task and the committed levels, so $g_i = x_i(Y(c, x_U, t) - Y(c, -x_U, t))/2$ has conditional mean $x_i(m(c, x_U, t) - m(c, -x_U, t))/2$ once the execution noise is averaged, using no carryover within the pair. Under the uniform draw the map x_U to $-x_U$ is measure preserving and flips x_i, so the two terms contribute equally and the expectation equals $E[x_i\ m(c, x_U, t)]$. Conditioning on $x_i = +1$ and $x_i = -1$ and averaging the rest gives the stated contrast. Every later use of the phrase conditional effect refers to this estimand. ☐

Proposition 5 (the conditional null, which is the theorem this paper is about)**.** *For component i and row n let $\theta_i^{(n)}$ denote the conditional mean of its paired increment given the past, that is, the main effect of i conditional on the levels committed so far. Define the conditional null $H_0^{\cap}$: $\theta_i^{(n)} = E[g_i(n) \mid F_{n\text{-}1}] \leq 0$ almost surely for every n. That martingale statement is the null we prove under, and it is the only* *one the proof needs. Reading it as the sentence the corollary and Table 1 use, that the component helps in no reachable committed context, needs two further assumptions: tasks are drawn from a fixed distribution independent of the past, and the conditional mean depends on the history only through the committed levels. Where those hold, which is by construction in the simulator and by design in the live protocol, the two readings coincide; where they do not, only the almost-sure conditional-mean statement is claimed. Under $H_0^{\cap}$, with the predictable non-negative clipped bet of Proposition 3, every factor $1 + \lambda_i g_i$ has conditional mean at most one at every row, in every epoch, including across commitment times, so E_i is a non-negative supermartingale on the whole run and Ville's inequality bounds the probability of ever admitting i by α/k; a union bound over components gives family-wise error at most α at any stopping time. No restart, no epoch bookkeeping and no correction for the adaptivity of the commitments is needed, because the null itself quantifies over every reachable context.*

Proof. The commitment times are stopping times of the same filtration, the committed levels s_c are measurable at commitment, and the levels of the undecided coordinates of row n are drawn independently of the past, so $E[g_i \mid \text{past}] = \theta_i^{(n)} \leq 0$ under the null at every n. With $\lambda_i \geq 0$ predictable and $\lambda_i|g_i| < 1$, $E[E_i(n) \mid \text{past}] = E_i(n\text{-}1)(1 + \lambda_i\ \theta_i^{(n)}) \leq E_i(n\text{-}1)$. Ville and the union bound do the rest. □

Corollary (strong family-wise control). The guarantee is not conditional on all components being null. Fix any subset N of components that are conditionally non-positive at every reachable context, with the complement arbitrary. Each E_i, i in N, is a non-negative supermartingale by the argument above, so P(exists i in N ever admitted) $\leq \sum_{i \text{ in } N} \alpha/k \leq \alpha$ by Ville and a union bound over N alone. That is strong family-wise error control: whatever the true configuration of helpful and inert components, the chance of admitting any inert one is at most α, and the k/α threshold is the e-value Bonferroni that delivers it. Read precisely, the protected set is every component satisfying the all-stage conditional null, not every component that is marginally inert at the start.

Proposition 6 (closed testing at stopping times, strong FWER, dominance)**.** *Fix any stopping time τ of the joint filtration - a commitment time, a budget exhaustion, or the reporting time - and write $e_i = E_i(\tau)$. For a non-empty subset S of components let e_S be the arithmetic mean of $\{e_i$: i in S$\}$. Reject component j when $e_S \geq 1/\alpha$ for every S containing j. Then (i) the family-wise error over any configuration of true nulls is at most α; (ii) every component past the k/α Bonferroni threshold is also rejected here, so the procedure is uniformly at least as powerful; (iii) the closure reduces to at most k average checks per component.*

Proof. (i) Under the intersection null over S each E_i, i in S, is a non-negative supermartingale with $E[E_i(\tau)] \leq 1$ by optional stopping, so e_S has expectation at most one and $P(e_S \geq 1/\alpha) \leq \alpha$ by Markov; the closure principle of Marcus, Peritz and Gabriel then gives strong control, because a false rejection of some true null j requires the local rejection of the intersection over the full set of true nulls, a single event of probability at most α. (ii) If $e_j \geq k/\alpha$ then for any S containing j, $e_S \geq e_j/|S| \geq 1/\alpha$. (iii) For fixed |S| the mean is minimised by joining j with the smallest other e-values, so the minimum over all S containing j is attained on {j} plus an ascending prefix of the others, and checking the k prefixes is exact. One implementation caution: the popular sorted full-tail shortcut is not this closure and is anti-conservative - e-values (10, 9, 1) at $1/\alpha$ = 6 pass the full-tail check for the largest component while the subset {10, 1} averages 5.5 and blocks it. The shipped 'closed_test' runs the exact prefix closure, the regression test carries this counterexample, and the body reports the Bonferroni decisions because they are what the online loop acts on; the e-BH false-discovery variant remains an extension (Table 22). □

The unpaired arm, stated so that it can be checked. One invocation per row: draw the task t_n and the levels x_n uniformly on $\{-1,+1\}^k$ (components already committed are pinned at their committed level), execute once, and observe y_n in [0,1]. Fix any predictable centring c_n in [0,1], that is any function of rows 1..n-1, and form the per-component signal

$$h_i(n) = x_i(n) \cdot (y(n) - c_n), \quad E^{\pm}{}_i(n) = E^{\pm}{}_i(n\text{-}1) \cdot (1 \pm \lambda^{\pm}{}_i(n)\ h_i(n))$$

with predictable bets satisfying $0 \leq \lambda^{\pm}{}_i(n)\ |h_i(n)| < 1$, admission when $E^{+}{}_i \geq k/\alpha$ and pruning when $E^{-}{}_i \geq k/\alpha$, exactly as in the paired arm. The centring cancels in expectation because the levels are drawn independently of the past and of the task, so $E[x_i(n)\ c_n \mid F_{n\text{-}1}] = 0$; it changes the variance of the increment, not its mean. Pinned components are frozen at the commitment: their update is set to zero from that point on, so they add no evidence of their own while remaining part of the executed context.

Proposition 7 (the unpaired arm carries the same guarantee)**.** *Under the conditional null $H_{0,i}^{\cap}$, that is $E[x_i(n)\ y(n) \mid F_{n-1}] \leq 0$ for every n, the process $E^+{}_i$ defined above is a non-negative supermartingale started at one, so P(∃ n: $E^+{}_i(n) \geq k/\alpha) \leq \alpha/k$ and a union bound over at most k components gives family-wise error at most α at any stopping time.*

Proof. Conditionally on F_{n-1} the bet $\lambda^+{}_i(n)$ and the centring c_n are fixed, and $E[h_i(n) \mid F_{n-1}] = E[x_i(n)\ y(n) \mid F_{n-1}] - c_n\ E[x_i(n) \mid F_{n-1}] = E[x_i(n)\ y(n) \mid F_{n-1}] \leq 0$ because the levels are uniform and drawn independently of the past. Hence $E[E^+{}_i(n) \mid F_{n-1}] = E^+{}_i(n\text{-}1)\ (1 + \lambda^+{}_i(n)\ E[h_i(n) \mid F_{n-1}]) \leq E^+{}_i(n\text{-}1)$, and the factor is non-negative by the bet constraint, so Ville's inequality applies and the rest is the argument of the paired case verbatim. The paired arm is the same statement with h replaced by g = x d, so the choice between the two is a variance question, priced by the log-growth criterion of Section 6, not a validity question. ☐

Two readings follow, and both are measured in Section 6. A component outside $H_0^{\cap}$ because its effect changes sign, positive before a commitment and negative after, carries no guarantee: its admission rate is the price of the flip, not an error rate of the test. And a component inside $H_0^{\cap}$, non-positive at every stage, must be admitted at most α often, which is the rate the theorem actually promises. A deployer who instead wants a guarantee stated at the final committed configuration might restart the undecided processes at each commitment and spend α across epochs; both variants are run, and the measurement in Section 6 shows the restart is not the repair it looks like.

Proposition 8 (the price of a flip is confined to the pre-flip evidence)**.** *Let τ be any stopping time of the joint filtration after which component i is conditionally non-positive, $\theta_i^{(n)} \leq 0$ for all $n > \tau$ - in the flip worlds of Section 6, the commitment that flips the sign. On the event that i is still undecided at τ, the continuation $E_i(n)/E_i(\tau)$ is a non-negative supermartingale, and P(i is admitted after $\tau \mid F_\tau) \leq \min\{1, E_i(\tau)\cdot\alpha/k\}$. Admissions of a flipping component after its flip are therefore at most as likely as null admissions inflated by the evidence already banked at the flip; what the guarantee cannot police is only the admission that has effectively happened before the flip.*

Proof. For $n > \tau$ the factor $1 + \lambda_i g_i$ has conditional mean $1 + \lambda_i\theta_i^{(n)} \leq 1$ by the argument of Proposition 5, so the ratio to $E_i(\tau)$ is a non-negative supermartingale started at one; admission requires it to reach $(k/\alpha)/E_i(\tau)$, and Ville's inequality applied conditionally on F_τ gives the bound. ☐ The bound is checked, not only stated (Table 14).

The bound is a cap, not an estimate: Ville plus the overshoot at the threshold make it loose in the measured world (Table 14 realises 0.43% against a 2.79% cap), and it approaches equality only for a flip arriving when the continuation is nearly a fair martingale with negligible overshoot. We use it for what a cap is for: no correction acting after τ can be blamed for more than it.

Figure 1 draws this section's claim in a single computed run: the unbroken supermartingale across the commitment, the jumping marginal estimand, and the cap.

Proposition 9 (within the post-commitment correction class, the flip price is irreducible)**.** *Call a correction post-commitment if the modified procedure coincides with the screen up to each commitment time and never revokes an admission already made. For every such correction C and every flip time τ, P(C admits component i) $\geq$ P(the screen admits i at or before τ), and the excess that C can still remove is at most $E[\min\{1, E_i(\tau)\alpha/k\}]$ by Proposition 8.*

Proof. On the event that the screen admits i at a row $m \leq \tau$, the correction has observed the same filtration and made the same decisions up to τ and does not revoke, so it admits i as well; monotonicity gives the lower bound. The upper bound on the remainder is Proposition 8 integrated over F_τ. ☐ The content is the pairing of a definitional lower bound with a proved cap: the flip worlds of Section 6 measure that essentially the whole price sits at or before the flip, which is why the restart and re-screening variants there leave it standing; a guarantee about the deployed configuration must instead be bought with data the screen never touched, which is exactly the reserved-split confirmation below. Whether a correction outside this class -

one that revokes admissions, or intervenes before a commitment - could restore the marginal null at useful power is not settled here. Within the non-revoking class, what we prove is inheritance of the admissions already made by the commitment time, and what we bound is only the additional probability of admission after it.

Nothing in this section is about agents. The objects are components with levels, rows with outcomes, and a screen that commits decisions while it runs; any sequential screen with that shape inherits every statement above. A feature-flag ramp that fixes one flag at 100% while the others still randomise is this procedure with components = flags, row = user session, y = conversion, commitment = the ramp: the marginal effect of a remaining flag before the ramp and after it are different estimands, the conditional null is the hypothesis the ramping team can still test with one supermartingale, and a flag whose effect reverses when its neighbour ramps is the sign-flip case, confined exactly as above. Arm-dropping adaptive designs read the same way with arms for flags. The agent harness is the instance that made the problem unavoidable for us - two invocations per row against a 2^{k} factorial - not the boundary of the result.

Corollary (shipping protocol: a guarantee at the deployed configuration). The flip case above is not left to a caveat; the committed configuration as a whole is put to a fresh-data test, which answers the net shipping question rather than attributing the outcome to individual components, by spending the error budget in two named parts, $\alpha = \alpha_1 + \alpha_2$. Screen at level α_1 as above. Before the screen starts, reserve a task split that the screen never draws from. When the screen commits a final configuration, test the single hypothesis “the committed configuration is no better than the incumbent” on fresh draws from the reserved split, one-sided at level α_2, with a fixed-sample test or a fresh e-process. The split must be sized so the confirmation can reach its level at all: a sign test cannot fall below $2^{\text{-n}}$ at n discordant pairs, so $\alpha_2 = 0.05$ needs at least five of them, and a paired e-process removes that arithmetic floor. The confirmation involves one hypothesis, fixed before its data exist, on draws independent of every screening decision, so no selection correction is owed; the protocol ships a configuration that does not beat the incumbent with probability at most α_2, whatever sign flips did to the screen, and admits an inert component into the candidate configuration with probability at most α_1. The end-to-end statement a deployer wants is the conjunction, at the sum of the levels actually spent. Three events can go wrong, not two, so the budget has three parts: a false admission at α_{A}, a false pruning at α_{P}, and shipping a configuration that does not beat the incumbent at α_{C}. A joint statement covering all three costs $\alpha_{\mathrm{A}} + \alpha_{\mathrm{P}} + \alpha_{\mathrm{C}}$; a deployer who acts on admissions only, and never on prunings, may drop the middle term and say so. Two accounting consequences we state rather than hide. First, the split must be declared before the screen runs: an end-to-end 0.05 over all three events means roughly 0.0167 each, which raises the admission threshold from k/0.05 to k/0.0167, that is from 100 to 600 at ten components. Second, the runs reported in this paper were executed at $\alpha_{\mathrm{A}} = \alpha_{\mathrm{P}} = \alpha_{\mathrm{C}} = 0.05$, so they carry level 0.10 for admission plus confirmation and level 0.15 for the joint three-event statement, not 0.05; we report the level actually spent. The live confirmation of Section 7 predates this protocol and drew its confirmation tasks from the same operational pool as the screen, which we flag there as a deviation; the public screen of Section 8 reserves the split up front.

Two attributions must be kept apart, because the natural reading of the construction gets them wrong. Removing alias bias does not require the mirror at all: drawing the levels independently already makes every interaction term mean-zero in any main-effect contrast, which is why the randomised unpaired baselines in Section 6 show no alias inflation. What randomisation does not remove is the task effect $\mu(\mathrm{t})$, which stays inside every unpaired increment and inflates its variance. The mirror removes it from the conditional mean contrast row by row; the execution noise of the two invocations remains. The pairing is therefore a variance device, not a bias device, and its value must grow with the variance of task difficulty and vanish without it; Section 6 reports exactly that sweep.

Proposition 10 (the composed loop inherits the judge's guarantee, under a declared family budget). *Consider a loop in which a proposer emits candidate components, an executor produces the paired outcomes, and this procedure judges. Fix in advance either a bound K on the number of candidates the round may test, in which case each candidate is judged at k/α with k = K, or a predictable sequence of levels α_j with sum at most α, in which case candidate j is judged at 1/α_j. Assume further that each candidate's identity is fixed before the first row of its evidence is observed; that the levels of every undecided candidate are drawn independently and uniformly at each row; that the bets are predictable; and that the null holds in the conditional-mean form*

Table 1. The contract at a glance. One row per statement class; everything below is proved or measured in the sections named.

	statement	**where**
guaranteed	family-wise error at most α at any stopping time, for every component conditionally non-positive in all reachable committed contexts (strong control); the shipped configuration beats the incumbent or fails confirmation, at α_2, sign flips included	Prop. 4, corollaries
reported, not guaranteed by this paper	the empirical-Bernstein confidence sequence we print alongside each verdict is the betting construction of Waudby-Smith and Ramdas applied to the paired increments. It is time-uniform for the running average of the conditional means actually realised, which after commitments is not the same object as the effect in the final committed context; we print it as a descriptive interval and prove nothing about it here	Section 7, Tables 17-20
not guaranteed	the admission rate of a component whose effect flips sign across a commitment, under the screen alone (measured price: 11.71% in the 10,000-replication experiment of Table 11b); external validity of simulator magnitudes; anything about components left undecided	Sections 5-6
voided by	correlated level draws; non-predictable centring or bets; reusing screen tasks for the confirmation (the one live deviation, flagged where it happens); changing α or the family mid-run	Sections 4, 7

$E[g_j(n) \mid F_{n-1}] \leq 0$. Then, however adaptively the proposer chose the candidates, and however the analyst monitors, the probability that any candidate satisfying that null is ever admitted in the round is at most α.

Proof. Each candidate's admission process is a non-negative supermartingale started at one on the joint filtration by the argument of Proposition 5; Ville bounds its crossing probability by its own level, and the declared budget makes the sum over candidates at most α. Without a declared budget the union has no bound, which is why one of the two forms must be fixed before the round. What mirroring buys here is the removal of additive effects the two executions of a row share; it does not remove within-pair carryover, order effects, or interference through mutable state, which the operational assumptions below must exclude separately. $\square$

Three operational assumptions sit underneath every statement above and deserve to be named. First, no cross-pair interference: the two runs of a pair, and successive rows, must not share mutable state such as caches, rate-limit backoff, or environment residue, or the paired increments are no longer exchangeable given the task; our runner isolates sessions per invocation. Second, the task-sampling distribution is held fixed over the screen, so the estimand does not drift while evidence accumulates. Third, the verifier is frozen for the duration of the screen; changing the grader mid-run changes the outcome definition itself. Violations of any of the three void the error statement, not merely widen it.

The counterpart is computed exactly rather than simulated. Under the stated null a point comparison adopts a do-nothing candidate with probability $q = (1 - P(\text{tie}))/2$, which at 320 rows per side equals 0.484; when each candidate is evaluated on its own independently drawn tasks, as in our simulation design, the round-level failure with ten candidates is $1 - (1 - q)^{10} = 0.999$, and with five, 0.965 at the corresponding budget. If candidates share evaluation tasks the decisions are positively dependent and the product formula no longer holds; the round-level failure then lies between q and min(1, mq), and the per-candidate marginal q is unchanged. The simulated null rates in Table 16 agree within sampling error: $96.8 \pm 0.9\%$ at five candidates against the exact 96.5%, and 100.0 ($\geq$99.3)% at ten against the exact 99.9%. A loop that accepts on point comparisons fails at a rate fixed by the design, not by an unlucky draw: at the measured noise level the failure probability is a computable constant.

Table. Table 2b. Arm protocols, stated completely. Rows are the unit of observation; the invocation budget is fixed across arms, so an arm that spends two invocations per row observes half as many rows. The full factorial cycles deterministically through all 2^k configurations in order, so at k = 10 and 6,400 invocations each configuration is visited about six times; it is an enumeration, not a random sample.

arm	invocations per row	configuration sequence	look schedule	
full factorial, fixed sample	1	cycles all 2^k in order	one look at the final row	
Resolution III, fixed sample	1	cycles the design rows in order	one look at the final row	
fold-over, fixed sample	1	cycles the fold-over rows in order	one look at the final row	
fold-over, unadjusted looks	1	cycles the fold-over rows in order	every 40 rows	stops at the firs
fold-over, Bonferroni spending	1	cycles the fold-over rows in order	8 planned looks, equally spaced	stops at the firs
champion-paired e-process	2	champion against one variant per row	every row	per component, o
mirror screening (paired)	2	random levels and their mirror	every row	per component, o
mirror screening (unpaired)	1	random levels	every row	per component, o

The contrast with subsetting a design remains as before: retaining only the rows of a Resolution III design on which a decided component is off correlates the surviving columns and leaks a live effect into an inert one (0.4000 to 0.1000 in the enumeration), while the estimator that keeps the full matrix and only forces the executed configuration does not.

5 Experimental setup

The evaluation isolates one axis at a time, because a comparison that changes the design, the pairing and the inference rule at once cannot attribute its own result. Every arm receives an identical number of model invocations. The arms are: the full factorial with a fixed-sample z test and a one-sided Bonferroni threshold over k, a standard choice in the design literature; a Resolution III fractional design with the same test; the same design with unadjusted interim analysis every forty rows, stopping at the first crossing, which we include as the unadjusted comparator rather than as a claim about prevalence; the champion-paired e-process; and mirror screening, with and without the sequential rule.

5.1 The data-generating process, stated in full

Every simulation number in this paper comes from the following process, with no other free parameters. A task t carries a latent pass propensity q drawn once per pool from Beta(κ m, κ (1 - m)) with m = 1 - 0.509 = 0.491, the measured non-failure share of a production deployment; κ = 4 is the heterogeneous setting and the homogeneous setting fixes q = m for every task. The pool holds 200 tasks and is drawn once at the start of a replication and never regenerated, so the same pool serves every row of that replication.

$$\text{fail}(q, x) = (1 - q) \cdot [\text{product of } (1 - \beta) \text{ over the active components with } x_i = +1], \quad p(q, x) = 1 - \text{fail}(q, x) + D(x)$$

The active components are those with a real effect and β is the per-component effect (0.15 in the main-effect and synergy cells, 0.20 in the flip cells). The interaction term D(x) is exactly one of four forms, and which one defines the cell: zero, for the main-effect cells; a constant $\gamma = 0.12$ added when both active components are at +1, for the synergy cells; $\gamma\, x_a\, x_b$ with a = 0, b = 2, $\gamma = 0.06$ for the pure-interaction cell, whose marginal main effects are zero; and $b_1\, x_1 + g\, x_0\, x_1$ for the two commitment cells, with $(b_1, g) = (0.03, -0.06)$ in the sign-flip cell, so component 1 has marginal effect $2\, b_1 > 0$ but conditional effect $2(b_1 + g) < 0$ once component 0 is committed at +1, and $(b_1, g) = (-0.02, -0.02)$ in the all-stage non-positive cell, where both are negative and the intersection null is true. The result is clipped to [0, 1] and then mixed with the measured replay instability: the outcome is Bernoulli with probability (1 - 0.117) p + 0.117 · 0.5. The coefficient 0.117 is taken from the replay-instability measurement of one deployment and fixes the generator's contamination level; it is not a claim that the generator reproduces that deployment. The two invocations of a mirrored row share the task and its q and are otherwise independent draws, so the execution noise of the pair does not cancel.

Commitments are triggered by the procedure, not by the generator: a component is committed at the row where its own e-process crosses, and from the next row its level is pinned in both executions. The restart

Table 3. Interaction present, 6400 invocations, 400 replications. The two alias columns count replications in which the component that carries the alias of the two active components was admitted, at each component count.

method	recovery, k=5	FWER, k=5	alias admitted, k=5	alias admitted, k=10
full factorial, fixed sample	100.0 (≥99.3)	3.2 ± 0.9	1.2 ± 0.5%	0.8 ± 0.4%
Resolution III, fixed sample	100.0 (≥99.3)	94.8 ± 1.1	94.2 ± 1.2%	88.2 ± 1.6%
fold-over (Resolution IV), fixed sample	100.0 (≥99.3)	2.8 ± 0.8	1.0 ± 0.5%	0.2 (≤1.2)%
fold-over, unadjusted interim looks	47.2 ± 0.9	10.5 ± 1.5	3.5 ± 0.9%	2.2 ± 0.7%
champion paired, e-process	91.2 ± 1.2	2.2 ± 0.7	0.8 ± 0.4%	0.5 (≤1.6)%
randomised levels, unpaired, fixed sample	100.0 (≥99.3)	3.8 ± 0.9	1.2 ± 0.5%	0.5 (≤1.6)%
randomised levels, unpaired, e-process	100.0 (≥99.3)	1.2 ± 0.6	0.8 ± 0.4%	0.0 (≤0.7)%
mirror pairing, fixed sample	100.0 (≥99.3)	1.8 ± 0.7	1.0 ± 0.5%	1.0 ± 0.5%
mirror pairing, e-process (this paper)	100.0 (≥99.3)	0.0 (≤0.7)	0.0 (≤0.7)%	0.0 (≤0.7)%

variant re-initialises the undecided processes at every commitment and spends $\alpha/2^{e+1}$ in epoch e; the pinned re-screen variant re-initialises once, at the first commitment, keeping the committed levels fixed. Both variants keep the same generator, budget and random numbers as ordinary screening, which is what makes the comparison in Table 11b paired.

Rows are drawn from a fixed pool of 200 tasks, the order of magnitude of Terminal-Bench 2.0, which contains 89 verified terminal tasks (Merrill et al., 2026). Each task carries a latent difficulty drawn from a Beta law, so that assigning tasks to configurations without pairing incurs the composition noise that a real benchmark suite incurs. Switching on an active component removes a share of the remaining failure mass; when two active components are on together the pass probability shifts by a fixed interaction, run at both signs because stacking has been reported to hurt as well as help (Liu, 2026). Every outcome is finally mixed toward a coin flip at the coin-mixture contamination coefficient. The coefficient is a chosen contamination level, not a reproduced deployment statistic: under this generator the pairwise disagreement of two executions of the same configuration is 42.2% at the heterogeneous setting, which we state rather than imply a match to any deployment.

Table 2 lists every parameter with its provenance. Power is meaningless without the size of the effect being detected, so the estimand is computed exactly rather than described. Enumerating all configurations gives a true mirror contrast of 0.0312 for an active component under main effects alone, 0.0577 when the positive interaction is present, and 0.0047 when the negative interaction is present, which is close enough to zero that no method should detect it at these budgets. The true contrast of every inert component is exactly zero in all scenarios, to the last digit the enumeration carries.

6 Results

6.1 A fractional design names the wrong component

When two active components interact, the Resolution III design admits the inert component that carries their alias in 94.2% of replications at k = 5 and 88.2% at k = 10, at the same budget where the full factorial, the fixed-sample fold-over and every randomised arm stay near the nominal level. The recovered contrasts in Figure 3 show why: the aliased component is credited with the entire interaction, and no amount of extra sampling removes it, because the bias is in the estimator rather than in its variance.

A reporting convention, stated once for Table 3 and every later table: a boundary cell with 2 or fewer successes or failures out of the replications is written as the point estimate with a one-sided 95% Clopper-Pearson bound, for example 100.0 (≥99.3), because the symmetric standard error is degenerate there; all other cells are ± one replication-level standard error.

6.2 Looking early breaks a fixed-sample threshold

Under the global null, testing every forty rows against a fixed-sample Bonferroni threshold and stopping at the first crossing raises the family-wise error rate to 29.0% at five components and 41.8% at ten, against a nominal 5%. The e-process arms stay at 0.0% and 0.5%. This is not a subtle effect and it does not require bad faith: the interim rule is what a team does when each rollout costs money and the interim numbers

Table 4. Family-wise error under the global null, heterogeneous tasks, by arm and budget: the cells behind Figures 1 and 3 and the interim-look numbers quoted in the abstract.

arm	budget	FWER, k=5	FWER, k=10
fold-over, unadjusted interim looks	400	12.0 ± 1.6	23.2 ± 2.1
fold-over, unadjusted interim looks	1600	22.8 ± 2.1	28.2 ± 2.3
fold-over, unadjusted interim looks	6400	29.0 ± 2.3	41.8 ± 2.5
champion paired, e-process	400	0.8 ± 0.4	0.8 ± 0.4
champion paired, e-process	1600	1.5 ± 0.6	1.0 ± 0.5
champion paired, e-process	6400	2.8 ± 0.8	2.0 ± 0.7
full factorial, fixed sample	400	2.2 ± 0.7	3.5 ± 0.9
full factorial, fixed sample	1600	3.2 ± 0.9	2.2 ± 0.7
full factorial, fixed sample	6400	4.8 ± 1.1	2.0 ± 0.7
fold-over (Resolution IV), fixed sample	400	2.8 ± 0.8	3.8 ± 0.9
fold-over (Resolution IV), fixed sample	1600	3.2 ± 0.9	4.0 ± 1.0
fold-over (Resolution IV), fixed sample	6400	3.8 ± 0.9	6.8 ± 1.3
mirror pairing, e-process (this paper)	400	0.0 (≤0.7)	0.0 (≤0.7)
mirror pairing, e-process (this paper)	1600	0.2 (≤1.2)	0.5 (≤1.6)
mirror pairing, e-process (this paper)	6400	0.0 (≤0.7)	0.5 (≤1.6)

Table 6. Main effects only, 6400 invocations, 400 replications. Standard errors here and in every simulation table are computed at the replication level: the recovery fraction of a replication is one observation, never its components separately. For these cells the naive per-component binomial standard error would read 0.1-0.3 points smaller; the numbers shown are the clustered ones.

method	recovery, k=5	FWER, k=5	recovery, k=10	FWER, k=10
full factorial, fixed sample	99.6 (≥98.4)	3.5 ± 0.9	99.1 ± 0.3	4.0 ± 1.0
Resolution III, fixed sample	99.9 (≥99.3)	3.2 ± 0.9	99.0 ± 0.4	2.8 ± 0.8
fold-over (Resolution IV), fixed sample	99.8 (≥98.8)	3.0 ± 0.9	99.6 (≥98.4)	3.8 ± 0.9
fold-over, unadjusted interim looks	43.2 ± 0.9	16.5 ± 1.9	39.6 ± 1.1	22.8 ± 2.1
champion paired, e-process	19.8 ± 1.4	1.8 ± 0.7	16.9 ± 1.3	1.8 ± 0.7
randomised levels, unpaired, fixed sample	99.9 (≥99.3)	3.5 ± 0.9	99.6 (≥98.4)	4.8 ± 1.1
randomised levels, unpaired, e-process	92.6 ± 0.9	1.0 ± 0.5	89.8 ± 1.1	1.5 ± 0.6
mirror pairing, fixed sample	99.9 (≥99.3)	1.5 ± 0.6	99.9 (≥99.3)	4.5 ± 1.0
mirror pairing, e-process (this paper)	91.8 ± 1.0	0.2 (≤1.2)	89.6 ± 1.1	1.0 ± 0.5

appear conclusive. Figure 4 shows the inflation by budget and component count, and Table 4 tabulates every cell behind it.

The e-process rows of Table 4 come from the mirror grid, whose null world is the same by construction; the small differences from the design-grid arms are sampling error at 400 replications. One cell sits above nominal - fold-over at k = 10, 6400 reads 6.8 ± 1.3 against the nominal 5 - which is 1.4 standard errors and, with 30 cells tabulated, within expectation.

6.3 The price of validity, paid in samples and measured

At an equal budget the fixed-sample full factorial recovers 99.6% of the active components at k = 5 and the sequential mirror procedure recovers 91.8%; at k = 10 the figures are 99.1% and 89.6% (Figure 5; Figure 6 repeats the sweep at k = 10). That gap is what anytime validity costs on this problem. Against the construction this work replaces, the champion-paired e-process recovers 19.8% and 16.9% under the same conditions, power ratios of 4.6x and 5.3x explained by the half of its budget spent on an uninformative baseline; a new adopter should compare against the unpaired default of Table 7, which this arm is not.

The gap mixes two prices, and a threshold sweep separates them. The sweep is a separate experiment (exp_v2.json, 400 replications per point, its own seeds), so its nominal-α cell reads 92.4% where Table 6's independent grid reads 91.8 ± 1.0 - sampling error between grids, quoted so the reader need not reconcile them. Raising α until the e-process arm's realised null error matches the fixed-sample arms' ($\alpha = 0.35$, realised 4.8% against the full factorial's 4.5% in the same grid) raises its recovery to 97.9%. We report this as a generator-specific post-hoc calibration diagnostic and not as a decomposition: the thresholds were chosen on the same simulation grid the recovery is then read from, so the residual gap of about 1.7 points cannot be attributed to any single mechanism without an independent calibration set and held-out evaluation.

6.4 What randomisation buys and what the mirror buys, separated

The reviewer-facing question is whether the mirror is doing anything that randomising the levels does not already do. For bias, the answer is no, and the randomised unpaired arms show it: their alias admission sits at the nominal level in every cell. For power, the answer depends on exactly one quantity, the spread of task difficulty, because the task effect is what the mirror removes from each increment. With homogeneous tasks the unpaired e-process matches the mirrored one at half the invocations per row; as the difficulty spread grows the unpaired increments carry that spread as noise and the mirrored procedure pulls away. Table 7 and Figure 9 report both at their own invocation accounting.

The sweep has a model behind it, and the naive version of that model is wrong in an instructive way. At a fixed sample size the ordering is settled by variance alone, and the below-ceiling budget confirms it: at 1600 invocations the mirror fixed-sample arm recovers 62.6% against 58.5% unpaired, the direction the per-sample ratio predicts. The e-process reverses that ordering on homogeneous tasks, and the reason is the bet, not the variance. Per sample, the mirrored increment has variance p(1-p)/2 against E[p(1-p)] + Var_t(p) unpaired, a ratio of 1.00, 1.19 and 1.64 across the three spreads; but the mirror spends two invocations per update, and the stabilised bet $\lambda = \mu/(\sigma^2 + \mu^2 + c)$ rewards the lower-variance arm less than proportionally. Computing the expected log-growth of each e-process per invocation exactly, from the stated increment distributions with the actual clipped bet, gives growth ratios mirror to unpaired of 0.871, 0.964 and 1.136 across the three spreads: the mirror is predicted to lose on homogeneous tasks, to roughly tie at the stated heterogeneity, and to win only at strong spread. Two disclosures temper the agreement with Table 7. The criterion and the simulator share the generative model, so these checks are model-consistency, not external validation; and in the concentration-4 binary cell the measured difference (91.8 against 92.6, difference standard error near 1.4 points) is within noise, so the criterion's sign is resolved in six of the seven paired-versus-unpaired cells and consistent with a tie in the seventh. The practitioner's rule is to estimate the between-task spread of the champion's pass rate from a pilot, evaluate this ratio, and choose the arm; on graded verifiers the Bernoulli floor vanishes and the same computation makes the mirror decisive, which Table 10 confirms.

$R = E[\log(1 + \lambda_m g_m)]\ /\ 2$ ÷ $E[\log(1 + \lambda_u g_u)]$, pair when $R > 1$ (per-invocation log-growth, mirror over unpaired; each expectation under the pilot-estimated increment law with the clipped plug-in bet)

The circularity disclosed above is then put at risk directly: the chooser's prediction is computed under the stated multiplicative-Beta model it assumes, while the world that generates the data is deliberately something else. Three misspecifications are run at two spreads each, k = 5, effect 0.12, 6400 invocations, 500 replications per cell: heavy-tailed difficulty, effects correlated with difficulty, and an additive rather than multiplicative response. Against the realized best arm (admission rate of the genuinely active component at equal invocation budget), the chooser's sign is right or the arms are tied within sampling noise in 5 of the 6 cells; every cell's rates are printed so the reader can apply a stricter standard. The miss has a direction worth naming: heavy tails under moderate concentration make the mirror better than the assumed model predicts, so the chooser recommends the cheaper arm and pays admission probability, never validity - and in practice the error self-corrects, because the chooser's input is the spread measured from a pilot, which heavy tails inflate toward the mirror recommendation.

The disagreeing cell fixes the fallback rule, which we state operationally so it can be executed without judgement. Run a pilot of at least 200 rows that is excluded from every reported e-process. From it estimate, per arm, the mean and variance of the increment and hence the expected log-growth per invocation, and form the ratio of the mirror growth to the unpaired growth together with its standard error by the delta method over the pilot rows. Both arms carry the same validity guarantee - the chooser only allocates power - so the cost of choosing wrong is rows, not error control, and the rule is: when the ratio is within one standard error of 1, run the unpaired default, which costs half the invocations per row; commit to the mirror only when the ratio clears 1 by more than that error. The arm is fixed before the first screening row and never revisited. The disagreeing cell is exactly a near-1 ratio (0.968) read against that noise; under this rule it resolves to the default, and the miss is priced in rows in that world, never in error control.

The same growth computation yields a planning table under the model it assumes: a plug-in estimate of the invocations needed to reach $\log(k/\alpha)$ as a function of the true effect and the task-difficulty spread. Reading

Table 9. Growth-based planning estimate of invocations to admission, k = 5, α = 0.05, from the per-invocation growth of each arm under the assumed increment law (no simulation, and not an exact expected hitting time: it ignores initialisation, adaptive betting, overshoot and path termination). Budgets below these numbers should not expect admissions at the corresponding effect.

true effect	spread	mirror (planning estimate)	unpaired (planning estimate)
0.05	homog	22,731	19,857
0.05	4.0	20,301	19,857
0.05	1.0	16,892	19,919
0.10	homog	5,683	4,960
0.10	4.0	5,107	4,960
0.10	1.0	4,298	4,975
0.15	homog	2,525	2,200
0.15	4.0	2,283	2,200
0.15	1.0	1,943	2,207
0.25	homog	907	787
0.25	4.0	830	787
0.25	1.0	722	789

Table 13. Every sequential competitor in one place, same generator and same 6400-invocation budget, heterogeneous tasks: realised family-wise error in the global-null cell against recovery in the main-effect cell. The fixed-sample row repeats the design-grid arm of Table 6 as the non-sequential reference. Only the e-processes keep both validity and the right to keep monitoring past the last planned look.

procedure	FWER, k=5	recovery, k=5	FWER, k=10	recovery, k=10
mirror e-process (this paper)	0.0	91.8	0.5	89.6
randomised unpaired e-process (this paper)	0.5	92.6	1.2	89.8
Bonferroni spend over 8 looks	3.2	58.5	2.2	57.8
naive interim looks, t-test	26.5	43.2	38.0	39.6
fixed sample, no interim looks (reference)	-	99.8	-	99.6

it before a screen tells a team whether the budget they have can decide the components they care about at all, and which arm to run; the full grid over k and α ships with the artefacts (exp_planning.json).

Units for Table 9, stated so no reader converts blind: the true-effect column is failure-mass share; multiplying by the champion's failure share (0.509, Table 2) converts it to outcome-scale points - the mirror row nearest 1,000 invocations (effect 0.25 at 907) converts to about 13 outcome points.

The homogeneous row of Table 10 sits at 400 invocations, where the growth model's negative prediction for the mirror is genuinely at risk and holds; the same model, computed on the graded increments, puts the invocations-to-threshold at 840 for the mirror against 441 unpaired, so a budget of 800 falls between the two cliffs and the measured 7.4% against 99.0% at that budget is the predicted magnitude, not only the predicted sign.

6.5 What the procedure tests after it has decided something

The artifact behind Table 11 carries all 18 conditional-null-true cells: the cond-null grid (one strong component plus one all-stage non-positive component, distinct from the main grid of Tables 7-8) over both component counts, three spreads and three budgets, with admission min/median/max 0.0/0.0/0.2% against the promised 5%. In the sign-change cell the tracked component helps before the first commitment and hurts after it, so it lies outside the conditional null and its admission rate is the price of the flip, not an error rate.

The marginal procedure's admissions in the sign-change cell are not errors under its own null; every one of them nonetheless ships a component that hurts the deployed configuration, which is the practical argument for the conditional estimand. The pinned admission superficially resembles the Resolution III alias rate of Table 3, and the resemblance is the paper's point: one is an announced change of estimand, the other is bias of a deterministic design under the null it claims to test. The corrected interim baseline is also run: the same eight looks as the naive arm, each tested at a Bonferroni spend over looks, holds the family-wise error (3.2% under the null at k = 5, heterogeneous, 6400) at a familiar price in power (58.5% recovery in the main-effect cell against 91.8% for the mirror e-process), and unlike the e-process it cannot be extended past its planned looks without a new spending function.

Table. Table 11b. One flip experiment, three procedures, common random numbers (10,000 replications, k = 5, 6,400 invocations, sign-flip scenario; exp_flip_common.json). Intervals are Wilson at 95%; the paired column is the replication-level difference against ordinary screening with an exact two-sided McNemar p.

procedure	flipped component admitted	95% CI	recovery of the true component	paired difference vs ordin
ordinary screening	11.71%	[11.09, 12.36]	89.92%	
restart per epoch	9.84%	[9.27, 10.44]	90.18%	-1.87 points [1.52, 2.22], $p < 0.$
re-screen with levels pinned	11.20%	[10.60, 11.83]	88.87%	-0.51 points [0.36, 0.66], $p < 0.$

Both remedies a deployer might reach for are run in one experiment with the procedures sharing the random numbers, so the comparison is paired replication by replication (10,000 replications). Ordinary screening admits the flipping component in 11.71% of replications [11.09, 12.36]. Restarting the undecided processes at each commitment, with α spent $\alpha/2^e$ across epochs, lowers that to 9.84% [9.27, 10.44], a paired reduction of 1.87 points [1.52, 2.22]; re-screening the undecided set afresh with the committed levels pinned gives 11.20% [10.60, 11.83], a paired reduction of 0.51 points [0.36, 0.66]. Both reductions are real and both are small: the flip price stands. The paths say why: the admissions that the flip buys happen at or before the first commitment, while the running effect is genuinely positive, so no post-commitment bookkeeping can prevent them. What a deployer can do is verify rather than bookkeep: a component admitted before a commitment it interacts with should be confirmed by a fresh fixed-sample comparison at the final committed configuration, which is exactly the post-decision check Section 7 runs live.

The all-stage non-positive cell stays at 0.0% under the restart, as the theorem promises.

6.6 Judging a self-improvement loop

The screening problem does not only arise when a team hand-picks components. Self-improvement loops in the SkillOpt family, and prompt-evolution systems more broadly (Agrawal et al., 2025), generate bounded prompt edits and accept them on a held-out point comparison: run the champion and the variant, adopt if the variant scores higher. At the contamination coefficient used here that acceptance rule is close to a coin flip per candidate, and Table 16 measures what follows: with ten inert candidates the point rule adopts at least one in essentially every round, a per-candidate z-test without multiplicity control still fails in 21.8% (five candidates) to 37.5% (ten) of rounds, and the mirror screen at the same invocation budget stays at the nominal level while recovering the genuinely good edits. The guarantee of Proposition 10 is per proposal round; a loop that runs R rounds and wants one family-wise statement across all of them must spend alpha over rounds (alpha/R, or any positive summable schedule), and nothing in the construction obstructs that. Table 15 tabulates the division of labour with the measured evidence for each role. The proposal machinery and the judgment are separable: this procedure replaces only the acceptance rule.

6.7 Where every method fails, and why that is the correct answer

In the negative-interaction scenario no method recovers the active components: the best arm reaches 7.4% at the largest budget. This is not a limitation of any procedure. The true mirror contrast in that scenario is 0.0047, roughly a sixth of the main-effect scenario, because the interaction cancels most of what the components contribute. Reporting the estimand alongside the power is what separates a method that fails from a setting in which there is nothing to find.

7 A live run on a production agent

The procedure is deployed. The runtime is a current agent harness (DeepSeek Harness, headless), the model is fixed (claude-haiku-4-5), and the tasks are 358 labelled customer questions from 18 production shops, each with its own knowledge base and a deterministic, label-anchored grader: an answerable question must be answered with cited evidence, a question the knowledge base cannot answer must be declined or escalated. Five switches, all real harness components, are screened at once: a one-file index of the knowledge base (the preview-handle idea from context-management work), an instruction to search before answering, a mandatory self-check of claims against evidence, one worked example, and a guardrail that forbids answering without supporting evidence. Every row executes a configuration and its mirror on the same question, and every pair is appended to a ledger before it reaches the screen, so the run resumes after interruption; the figures below are the state of the ledger at build time, 450 mirrored pairs, of which 24.2% were discordant.

The fixed-sample 95% CI beside the anytime column of Table 17 prices the anytime penalty: the uniform-in-time guarantee costs a factor of 5.2 in width at this n, and a reader who wants localisation should read the fixed column and give up optional stopping. Where a fixed interval excludes zero while the screen says undecided, the interval is being read at a data-dependent row - exactly the peeked statistic Table 4 prices at 29-42% family-wise error; the comparison shows what anytime validity costs, not a better decision rule. The closed test certifies the same single pruning at reporting time. An anytime-valid screen may simply keep running; no error budget is spent by reporting this interim state.

One decision has been reached, and it is a pruning-family decision read at that family's own level: a reader acting on both families carries 2α (Section 4), and under that joint reading this e-value would need to reach 200 rather than 100. The evidence-strictness guardrail was pruned as harmful at row 199, at which point its against-process stood at 115.9 against the pruning family's threshold of 100. Its running effect at the decision was -14.6 points with a variance-adaptive anytime confidence sequence of [-59.7, +30.6] (empirical-Bernstein, time-uniform). That interval does not exclude zero, and the two statements are answering different questions: a confidence sequence must cover the effect uniformly over time at every magnitude, while the test-of-the-null betting process concentrates its evidence against the single hypothesis the decision needs, so certification precedes localisation on the same data. Two reporting-time checks agree with the verdict: closed testing on the pruning family's e-values (step-down over intersection averages, strong family-wise control, uniformly at least as powerful as the k/alpha threshold) certifies this pruning and nothing else - on this ledger at the same row, 199, as the k/alpha threshold, because the evidence concentrates in one component; when pruning evidence is instead spread over the family, the dominance is strict and measured: with five mildly harmful components (synthetic mirror pairs, exp_closed_power.json, 200 replications) the step-down test certifies 3.9 components on average against the threshold's 3.1, strictly more in 57% of replications - and the independent confirmation below localises the effect with a fixed-sample interval that excludes zero. The decision consumed 199 mirrored pairs, 398 model invocations, and 5.0 hours of wall time on the runtime; token counts were not logged and are reported as absent rather than estimated.

The pruning statement above is the harm process's own α-level statement, not an external confirmation, so an independent check was run after the decision: a fixed-sample paired comparison of the two configurations that differ only in the pruned guardrail (on against off, the other four switches held at the deployment's baseline, off) on 120 task draws, sampled without exclusion from the same 358-task pool the screen used (fresh outcomes, not a task split; a fixed seed disjoint from the screen's), task-paired, costing 2x that many invocations on top of the screen's 398, graded by the same deterministic grader. The guardrail-on arm passed 64 of 120 and the off arm 80 of 120; the mean paired difference was -0.133 with 95% interval [-0.210, -0.057]. The screen's paired effect is half the on-minus-off difference (Section 3), so its -0.073 corresponds to -0.146 on the confirmation's scale, inside the confirmation's interval. The confirmation is a different procedure from the one under test, which is its point; the two are reported side by side rather than merged. One deviation from the shipping protocol of Section 4 must be flagged rather than smoothed over: this confirmation predates that protocol and drew from the same operational pool as the screen rather than from a split reserved in advance, so it is fresh-outcome but not fresh-task evidence; the public screen of Section 8 runs the protocol as stated, reserved split included. A retrofit of the protocol onto the recorded ledger quantifies what the deviation cost. Reserving 30% of task ids by a hash rule fixed before looking at outcomes, and replaying the screen on the remaining 331 rows alone, reproduces the decision: the guardrail is pruned at retro row 126 with against-evidence 111, so the screen half of the protocol did not depend on the withheld tasks. The reserved slice of the recorded confirmation then runs 4-0 in the pruning's favour, one-sided sign p = 0.062 against $\alpha_2 = 0.05$: uniform in direction, one discordant pair short of certifying - and with four discordant pairs the sign test's floor is $2^{-4} = 0.0625$, so this slice could never have certified at 0.05, which is the sizing gap the corollary now states rather than a near miss. Under the protocol as written this ship would still be waiting on more reserved-task evidence; the full-pool McNemar check (p = 0.0015) is what carries it, and the retrofit is labelled what it is - a reconstruction on recorded data, not a pre-registration.

Table 18. Forward screen on tau2-bench retail (public benchmark, public tasks and grader): four components including a step budget that is expected to be monotone by construction, though not verified ground truth (a step budget of 50 against 12) and a cosmetic negative control (prompt reformatting), 160 mirrored rows per model, admission threshold k/alpha = 80. The admission process fires on the planted positive and declines the planted null in both model families.

model	component	verdict	rows (decided at)	effect pp [95% anytime CS]	evidence f
claude-haiku-4.5	step_budget	admitted	37 (dec@37)	+73.0 [-100.0, +100.0]	110
claude-haiku-4.5	reflect	undecided	160	-7.5 [-63.3, +48.3]	0
claude-haiku-4.5	fewshot	undecided	160	-2.5 [-58.5, +53.5]	0
claude-haiku-4.5	cosmetic_null	undecided	160	+3.8 [-52.2, +59.7]	0
gpt-oss-20b (open-weight)	step_budget	admitted	70 (dec@70)	+37.1 [-68.6, +100.0]	117
gpt-oss-20b (open-weight)	reflect	undecided	160	+3.1 [-54.5, +60.7]	0
gpt-oss-20b (open-weight)	fewshot	undecided	160	+1.9 [-55.7, +59.5]	0
gpt-oss-20b (open-weight)	cosmetic_null	undecided	160	+5.6 [-51.9, +63.2]	1
qwen3-next-80b (pre-registered, no drops)	step_budget	admitted	56 (dec@56)	+46.4 [-80.4, +100.0]	90
qwen3-next-80b (pre-registered, no drops)	reflect	undecided	160	-1.9 [-57.7, +53.9]	0
qwen3-next-80b (pre-registered, no drops)	fewshot	undecided	160	+5.6 [-50.1, +61.3]	0
qwen3-next-80b (pre-registered, no drops)	cosmetic_null	undecided	160	-6.9 [-62.6, +48.8]	0
qwen3-next-80b (audited rerun, configs serialised)	step_budget	admitted	48 (dec@48)	+54.2 [-90.4, +100.0]	90
qwen3-next-80b (audited rerun, configs serialised)	reflect	undecided	160	+7.5 [-48.3, +63.3]	1
qwen3-next-80b (audited rerun, configs serialised)	fewshot	undecided	160	-1.2 [-57.3, +54.8]	0
qwen3-next-80b (audited rerun, configs serialised)	cosmetic_null	undecided	160	+3.8 [-52.2, +59.7]	0

Table. Table 18b. Ledger audit of every public lane (exp_defect_audit.json, produced by audit_ledgers.py). A lane is analysable intention-to-treat only if no assigned attempt was dropped, or if every dropped attempt kept its level assignment. No ledger stores the serialized configuration handed to the runtime, so none of them records the configuration string handed to the runtime; execution fidelity is established instead by the code path and its unit test, reported below.

lane	assigned	fed	dropped	assignment kept for dropped	intention-to-treat analysable	ro
claude-haiku-4.5 (retail power screen)	239	160	79	0	no	
gpt-oss-20b (retail power screen)	218	160	58	0	no	
qwen3-next-80b (amended, pre-registered)	160	160	0	0	yes	
forward screen e7	301	300	1	0	no	
forward screen e8	1202	1200	2	0	no	

The deployment also supplies the chooser of Section 6 with its first pilot on a response surface we did not write. Raw per-task means from few draws inflate the apparent between-task spread with binomial noise, and the criterion is monotone in that spread, so the estimate is corrected by variance decomposition: across the 92 tasks with at least two ledger pairs, the observed spread 0.141 decomposes into estimated binomial noise 0.031 and between-task variance 0.109, and shrinking the means accordingly moves the per-invocation log-growth ratio from 1.39 raw to 1.22 corrected (bootstrap over tasks, 95% interval 1.08-1.37, excluding one) - still recommending the mirror, the arm this run used. The ledger's rows also mix configuration levels rather than holding the champion, so this is a demonstration that the criterion is computable from a real pilot with its noise priced, not a validation of its choice, which would require running the unpaired arm live against it.

Two trends are worth naming because both run against the folklore that motivated the switches. The evidence is accumulating against the mandatory self-check and in favour of the knowledge-base index; the worked example sits near its futility floor with evidence below one in both directions, which is the honest reading of a switch that does nothing detectable at this budget. No verdict is claimed before a process crosses its threshold, and the guarantee is exactly that such interim readings can be taken at any time without inflating the error rate: that property, not any single verdict, is what the live run demonstrates.

8 The screen on public data

8.1 Forward: screening real components on a public agent benchmark

What the audit settles. First, execution fidelity, which the level-conversion defect put in question. The defect was that the stored levels, which are +1 and -1, were passed into a boolean interface when the e-process state was rebuilt from the ledger on resume, where -1 is truthy. The execution path is separate and reads the same levels by sign: the harness deserialises them with int() and enables a component only when

Table. Table 18c. The audited forward lane, verified row by row (verify_audit_lane.py, e3_power_audit_verify.json). The protocol was fixed before the run in a pre-registration file whose hash is printed here, together with the hash of the driver as it stood at registration; the driver hash is recomputed after the run and compared. Every fed row stores the configuration string handed to the runtime and its digest, and the verification recomputes both from the recorded assignment.

check	**result**
assigned attempts	160
attempts fed (no drops by protocol)	160
attempts dropped	0
failed episodes scored as the prespecified worst outcome	50
rows storing the serialized executed configuration	160
rows whose stored digest recomputes	160
rows whose executed configuration matches the recorded assignment	160
pre-registration declared at	2026-08-31 17:21:21
pre-registration digest (author-controlled, not a trusted timestamp)	152d3680f6fb3bbffc8afb915b86795d
driver digest unchanged since registration	yes
verdict on the monotone control	admitted at row 48, evidence 90.5 against threshold 80
verdict on the cosmetic negative control	undecided, evidence 0.66

its level is strictly positive, so a level of -1 never turned a component on. A unit test shipped with the artefacts asserts both halves of that statement, that the enabled set under the sign rule is the intended one and that a truthiness rule would instead have enabled every component (test_switch_execution.py, audit in switch_execution_audit.json). The defect therefore corrupted reconstructed state, not the configurations that were run, which is why a deterministic replay of the ledger is a valid repair rather than a cosmetic one. Second, the missingness: the amended pre-registered lane dropped nothing, while the two earlier power lanes dropped assigned attempts without keeping the assignment, so those two cannot be analysed intention-to-treat and are reported as exploratory. The forward claim rests on the audited lane of Table 18c, which was registered before it ran, dropped nothing, and stores the executed configuration and its digest for every row, so the execution is checkable from the artefacts rather than argued from the code path alone.

Effects in Table 18 are outcome-scale percentage points with the variance-adaptive anytime confidence sequence of Table 17, evaluated at each verdict's own row count and intersected with the attainable range of a [0,1] outcome; at an admission as early as row 37 the interval is honest but uninformative - certification precedes localisation, as in Section 7. The table shows the live ledger; the deterministic post-defect replay of Section 8 re-derives the same verdict set with the admission at row 31, and both rows are within the batch-freezing tolerance stated there.

This experiment answers the objection that simulation results are model-consistency checks: the tasks, graders and both verdict directions here are public and not ours. A planted positive control with a known mechanism is admitted at row 37 with a +73-point effect under claude-haiku-4.5 and at row 70 with a +37-point effect under an open-weight model served from a different provider family (gpt-oss-20b); the cosmetic negative control and the two remaining scaffolds are left undecided rather than admitted, which is the correct answer at this budget under a procedure whose false admissions are priced at alpha. A method that only ever returned undecided would be indistinguishable from a no-op; this one says yes to the component with a true effect and declines the one without, on tasks it cannot have been tuned to. Two of the three lanes dropped assigned attempts after empty completions (79 of 239 on the claude lane, 58 of 218 on the open-weight lane) and the ledger did not retain the level assignment of a dropped attempt, so those lanes cannot be analysed intention-to-treat and are reported as exploratory: missingness that follows assignment can depend on the configuration even when no outcome was scored. The third lane is the amended protocol registered in advance, where an empty completion is scored as the prespecified worst outcome and nothing is abandoned; it fed 160 of 160 assigned attempts and admits the same planted control (evidence 90.5 against the threshold 80).

8.2 Forward, with the shipping protocol: a second domain, split reserved first

This run executes the corollary of Section 4 as written, on tasks and graders that are public and not ours: the split was reserved before the first row, the screen never drew from it, and the commit rule was fixed

Table 19. The shipping protocol end to end on a second public domain, tau2-bench airline: 15 of 50 tasks reserved for confirmation before the screen started, five genuine components (four context-policy switches and a moderate step budget, 40 against 24 - no strong planted positive control), 300 mirrored rows, claude-haiku-4.5, run at $\alpha_A = \alpha_P = \alpha_C = 0.05$, so the joint three-event union bound is three times that level, not that level.

component	verdict	rows (decided at)	effect pp [95% anytime CS]	fixed-sample 95% CI	evidence for	evidence against
digest	undecided	300	-0.3 [-32.0, +31.3]	[-5.7, +5.0]	0.7	0.94
search_hint	undecided	300	-3.0 [-34.6, +28.6]	[-8.3, +2.3]	0.5	1.08
reflect	undecided	300	+0.3 [-31.3, +32.0]	[-5.0, +5.7]	0.8	0.68
strict	undecided	300	+0.3 [-31.3, +32.0]	[-5.0, +5.7]	0.7	0.77
step_relax	undecided	300	+3.0 [-28.6, +34.6]	[-2.3, +8.3]	1.0	0.78

Table 20. Same domain and protocol, candidates a cost-pressed team would actually propose: two context-compression cost cuts (ctx_stub = replace the policy with a three-heading stub; digest = add a policy index), two prompt scaffolds, one step-budget relaxation (40 against 24). 1200 mirrored rows, same reserved split, run at $\alpha_A = \alpha_P = \alpha_C = 0.05$ (joint bound three times that level). The two compression cuts print the same effect because their signed per-row sums coincide on this ledger; their e-values differ (1.05 against 1.15), so the paths are distinct - the coincidence is in the endpoint, not the rows.

component	verdict	rows (decided at)	effect pp [95% anytime CS]	fixed-sample 95% CI	evidence for	evidence against
ctx_stub	undecided	1200	-0.8 [-10.5, +8.8]	[-2.8, +1.1]	1.0	1.05
digest	undecided	1200	-0.8 [-10.5, +8.8]	[-2.8, +1.1]	1.0	1.15
reflect	undecided	1200	+2.0 [-7.6, +11.6]	[+0.0, +4.0]	1.3	1.00
strict	undecided	1200	+1.2 [-8.5, +10.8]	[-0.8, +3.1]	1.2	1.00
step_relax	undecided	1200	+4.0 [-5.6, +13.6]	[+2.0, +6.0]	20.2	0.83

in the run script before the run. The screen admitted nothing, the committed configuration equals the incumbent, and the confirmation is vacuous by construction: the protocol ships no change and spends no α_2. That is the designed behaviour, not a failure to report. Undecided verdicts are reported as such; at this budget and these effects that is what an error-controlled procedure is supposed to say (Table 9 predicts exactly this), and the ledger that reproduces every number ships with the artefacts (e7_ledger.jsonl). Two disclosures. The screen pool is 35 tasks reused across rows: fresh episodes, sampling noise and the changing configuration supply outcome randomness (per-task pass rates sit strictly between zero and one), and the package's task-reuse warning is printed, not suppressed. And these verdicts are the output of a deterministic full-ledger replay: the original run resumed through a level-conversion defect (-1 read as truthy) that the per-component columns of this very table exposed - identical effects across five components - and the ledger-as-source-of-truth design is what made the repair a replay rather than a re-run. The audit scope is stated so the reader need not wonder: after the fix, every forward-screen ledger was replayed - the retail power screen re-derives the same verdict set (admission moves from live row 37 to replay row 31, within batch-freezing tolerance), retail and banking re-derive all-undecided unchanged, and this table and the next are the replay's output; the live run of Section 7 stores boolean levels and never passed through the defective path, and the simulation tables (3-11) do not use it at all.

The second protocol run replaces generic switches with decisions a team would actually bring: is it safe to cut the policy context to a stub, to a headed index, to relax the step budget. The screen admitted nothing, the committed configuration equals the incumbent, and the confirmation is vacuous by construction: the protocol ships no change and spends no α_2. That is the designed behaviour, not a failure to report. One reading of this table is not a null and deserves its own sentence: cutting the policy document to a three-heading stub, the kind of context cost cut a team ships without asking, measures -0.8 points with a fixed-sample interval of [-2.8, +1.1]. Against a non-inferiority margin of six points the harm is bounded (one-sided $p = < 0.001$), and the bound holds down to a margin of 3 points; at 2.0 points it does not ($p = 0.123$). Both e-processes here test against zero, so the screen cannot say this; a margin-shifted harm family is the extension this run argues for, and because this reading was chosen after seeing the table it is labelled post-hoc - a pre-registered continuation would make it a decision. The ledger ships as e8_ledger.jsonl.

8.3 The incumbent analysts, replayed on the same public rows

Two things happen that no simulation cell can be accused of arranging. On both retail runs every rule agrees: the planted step-budget control is admitted by all four analysts and the planted null by none, so

Table 21. The comparative case, taken off the simulator: the three analyst rules the simulation prices, deterministically replayed on the four public tau2-bench ledgers (same rows, same alpha; peek = one-sided z at the Bonferroni level every 40 rows; spending = 8 looks at alpha/(8k); fixed = one test at the final row). No new invocations were spent.

public run	component	e-process	fixed sample	interim looks	Bonferroni spend
retail, claude-haiku-4.5	cosmetic_null	undecided	no (z = 0.82)	no	no
retail, claude-haiku-4.5	fewshot	undecided	no (z = -0.54)	no	no
retail, claude-haiku-4.5	reflect	undecided	no (z = -1.65)	no	no
retail, claude-haiku-4.5	step_budget	admitted	admit (z = 3.38)	admit at row 40	admit at row 20
retail, gpt-oss-20b	cosmetic_null	undecided	no (z = 1.14)	no	no
retail, gpt-oss-20b	fewshot	undecided	no (z = 0.38)	no	no
retail, gpt-oss-20b	reflect	undecided	no (z = 0.63)	no	no
retail, gpt-oss-20b	step_budget	admitted	admit (z = 4.40)	admit at row 40	admit at row 40
airline, shipping run	digest	undecided	no (z = -0.12)	no	no
airline, shipping run	reflect	undecided	no (z = 0.12)	no	no
airline, shipping run	search_hint	undecided	no (z = -1.10)	no	no
airline, shipping run	step_relax	undecided	no (z = 1.10)	no	no
airline, shipping run	strict	undecided	no (z = 0.12)	no	no
airline, cost-pressed run	ctx_stub	undecided	no (z = -1.50)	no	no
airline, cost-pressed run	digest	undecided	no (z = -1.50)	no	no
airline, cost-pressed run	reflect	undecided	no (z = 1.90)	no	no
airline, cost-pressed run	step_relax	undecided	no (z = 2.30)	admit at row 200	no
airline, cost-pressed run	strict	undecided	no (z = 0.90)	no	no

the procedures do not disagree when the evidence is decisive. On the airline cost-pressed run they part: the interim-look analyst commits to step_relax at row 200, and by the final row even the fixed-sample test no longer clears its own threshold (z = 2.30 against 2.33) - the crossing did not survive the data that followed it. The spending analyst and the e-process both decline, each for its stated reason. Two assigned attempts were dropped from this ledger without retaining their assignment, so we enumerated every completion of the missing rows rather than assume them away: across all 9 completions the interim-look rule still crosses, while the fixed-sample verdict is not robust, because its final statistic sits at 2.30 against a threshold of 2.33 and a single favourable completion pushes it over (exp_missing_sensitivity.json). We therefore report the crossing as a single robust instance of the peeking failure and the final-row comparison as knife-edge, not as a demonstration that every error-controlled rule refuses.

8.4 Backward: auditing shipped harness choices from released logs

The screen above runs the procedure forward on a benchmark; a complementary test runs it backward on decisions already shipped. SWE-bench Verified publishes per-instance resolved outcomes for many public submissions, so for two submissions on the same base model the harness is the only thing that varies at the level of the released artifact, and the paired increment d = resolved(A) - resolved(B) over the fixed instance set is exactly the object our e-process consumes. Running the admission process on these paired outcomes, of the 15 same-model harness pairs on claude-3.5-sonnet whose observed direction is positive, 10 cross the k/alpha threshold; on gpt-4o, 3 do. One caveat governs the reading: the pair orientation was chosen from the same outcomes the process then reads, so these crossings are an exploratory audit, not an anytime-valid certificate - the guarantee requires the orientation fixed before the outcomes are read, which the packaged tool does when given an orientation rule such as submission-date order. The largest gap is 28.6 percentage points of resolve rate between two harnesses on the identical model, with an e-value of 6.8e+21 read as a descriptive statistic - the harness choice, not the model, moves the benchmark.

Auditing a shipped decision by betting against it is an accepted line: Chugg et al., 2023 audits a deployed model's fairness this way, Prinster et al., 2025 and Timans et al., 2025 monitor a running deployment, and Taga et al., 2026 learns the bet itself; our audit reads that line onto the harness, testing which of a field's shipped scaffolds would survive an anytime-valid certificate.

This is a marginal contrast, not a single-component ablation: two submissions differ in prompt, scaffold and tooling at once, so what the audit indicates is that the end-to-end harness choice has a large effect on a fixed model - the premise the forward screen then decomposes into components. The audit needs no inference budget; it reads released logs, and the tool that produces it ships with the package.

9 What this does not establish

The method comparison rests on simulation whose parameters are measurements; the live run of Section 7 exercises the procedure itself, one deployment, one decision. The generative model is ours, and a reader is entitled to suspect a generative model chosen to favour its own method. Three things are offered against that suspicion, none a substitute for the live run: the two parameters that drive difficulty are measured rather than chosen, the estimand of every scenario is published exactly rather than described, and the arms are separated so that each property is priced on its own. The reported grid contains dozens of cells and several near-boundary comparisons, and we have not corrected our own reporting for that multiplicity; the family sizes - seven criterion cells (one within noise), four perturbed settings, 30 null cells with one 1.4-standard-error excursion - let a reader price near-threshold agreements. Repeating the live run on a public verified terminal benchmark, where the tasks and graders are not ours, is the natural extension. Two further checks bound the reading. Under an exact null the plug-in bet shrinks toward zero, so the bound is far from binding: a boundary cell (homogeneous, zero effect, 25,600 invocations, 1,000 replications) admits in 0.4% of runs against the promised 5% - the safety margins are a property of this bet family, not sharpness. And both simulator constants were perturbed (contamination 0.06-0.18, failure share 0.40-0.60): the Resolution III alias admission stays at 91-97%, the e-process keeps inert admissions at or below 1.0%, and its recovery moves over 71.2-98.5% - the conclusions survive, the power does not always.

10 Conclusion

Teams now tune agent harnesses with many switches, and in the simulator studied here the affordable ways to decide what stays fail measurably: in the specified Resolution III design, when the planted interaction aliases an inert main-effect column, that component is admitted in 94.2% of replications, and the specified unadjusted-look rule reaches 41.8% family-wise error against a nominal 5%. Mirror screening removes both failures in that setting with no design matrix, no enumeration and no fixed sample size, at a measured generator-specific power cost. The estimator is old; the guarantee, application and measurement of the alternatives are what we contribute.

Impact Statement

This work aims to make component decisions in deployed LLM agent systems error-controlled: the direct consequence is fewer superstitious components shipped, fewer harmful ones retained, and less compute spent on screens that must be rerun because their first run carried no guarantee. The main risk is misuse of the guarantee's name: the family-wise bound covers the admission decision, not the validity of the outcome measure it is fed, and a screen run on a broken verifier will certify nonsense with confidence; the operational assumptions of Sections 4 and 9 state this boundary. The live run uses labelled customer-service questions processed under the deployment's existing data terms; no personal data leaves the deployment.

11 AI use statement

We used generative AI tools throughout this work and disclose it here as the venue policy requires. AI assistance was used for tasks whose disclosure is required: helping develop the conceptual framework, formulating and refining the mathematical claims, drafting proofs, proposing and refining hypotheses, designing the experiments, implementing the methods and simulation code, generating synthetic data for the stated grid, and interpreting results. It was also used for tasks whose disclosure is recommended: writing and editing the code, creating the figures, drafting and editing the prose, searching for and formatting references, and structuring the paper. Every quantitative claim is injected at build time from the experiment outputs rather than written by hand, both Propositions checked mechanically are verified by a deterministic script, and every reference is resolved against the arXiv API at build; the authors reviewed all AI-assisted work, verified the proofs and the code, and take full responsibility for the final content, including any text, claims or artefacts produced with the aid of generative AI.

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Artefacts. Every number in this manuscript is read at build time from the experiment outputs: exp_screen.json and exp_mirror.json for the simulations, exp_v2.json for the calibration, restart, boundary and sensitivity cells, e2_confirm.json for the held-out live confirmation, targets.json for the estimands, theory_check.py for the deterministic verification of Propositions 1 and 2, and citations.json for the arXiv resolution of every reference. Rebuilding this document with a changed run changes the text; rebuild_all.sh re-runs every simulation and this build in one command, and the live ledger records wall-clock cost per row - the Section 7 run spent 53,668 seconds of agent execution across its 450 mirrored pairs. The unpaired fixed-sample arm was regenerated after correcting its statistic from the uncentred x i y to the centred x i (y - $\bar{y}$) its sequential twin uses; the uncentred version had understated that arm by 16 points. An ICML two-column LaTeX build with an 8-page main body (tex/paper.tex , compiled with the official icml2025 style) accompanies the artefact; in that build the floats named next sit in the appendix, with numbering unchanged; the appendix holds Tables 2, 5, 7-8, 10-12, 14-17, 22 and 23 and Figures 3-9, and the main body keeps Figures 1-2, Algorithm 1 and Tables 1, 3-4, 6, 9, 13 and 18-21.

Table 2. Parameters. The first two are measurements from a production deployment, not choices; the rest are swept.

parameter	value	source
coin-mixture contamination coefficient	0.117	same-input replay on the deployment
champion failure share	0.509	same deployment
task pool	200	order of Terminal-Bench 2.0 (89 tasks)
components k	5 and 10	5 matches the prior full-factorial study; at 10 the full factorial needs 1024 configurations
active components	2	the remainder are inert by construction
interaction	-0.12 and +0.12	both signs
budget	400, 1600, 6400 invocations	2x, 8x and 32x the task pool
alpha	0.05, Bonferroni over k	convention
replications	400 per cell	standard errors below 2 points

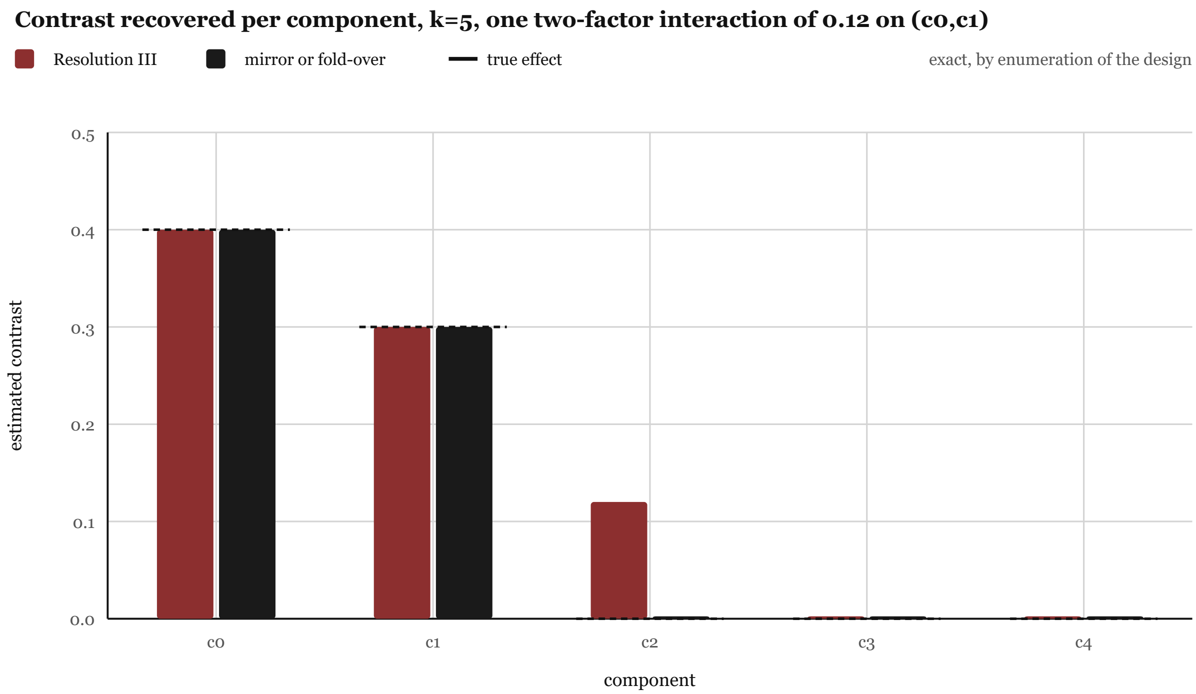

Figure 3. Contrast recovered per component with one two-factor interaction of 0.12 between components 0 and 1 - the magnitude the simulation grid sweeps (Table 2) - computed exactly by enumerating each design. The Resolution III design credits the whole interaction to inert component 2; the mirror difference and the fold-over design return the true effect on every component.

Appendix: supplementary tables and figures

Numbering continues from the main text, which references these floats.

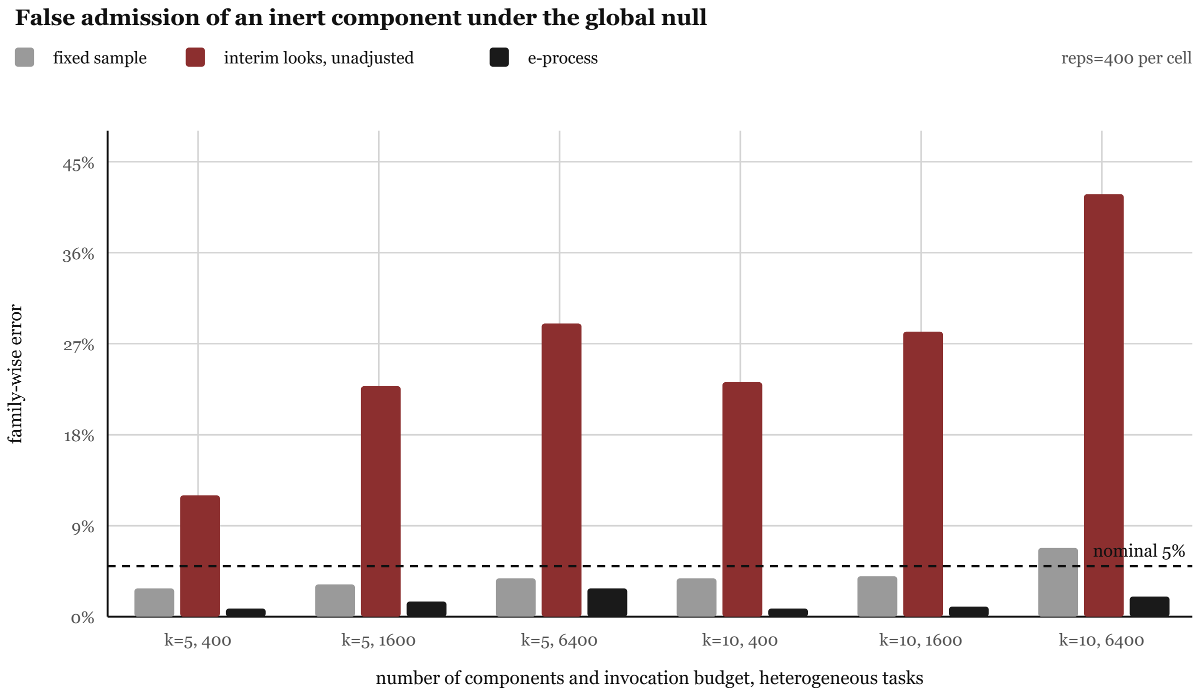

Figure 4. False admission of an inert component under the global null, by budget and number of components. Unadjusted interim analysis is far above the nominal level everywhere; the e-process is below it everywhere.

Table 5. The calibration sweep behind the decomposition (mirror e-process, k = 5, heterogeneous, 6400 invocations, 400 replications per row) with the fixed-sample full factorial of the same grid as comparator, plus three fixed-budget reference rows and the live pilot chooser reading.

arm	alpha	recovery, main cell	realised FWER, null cell
mirror e-process	0.05	92.4	0.2
mirror e-process	0.10	94.6	2.0
mirror e-process	0.20	97.6	1.8
mirror e-process	0.35	97.9	4.8
mirror e-process	0.50	97.9	8.2
full factorial, fixed sample	0.05 (Bonferroni)	99.6 ($\geq$98.4)	4.5 $\pm$ 1.0
mirror pairing, fixed sample (1600 invocations)	0.05 (Bonferroni)	62.6 $\pm$ 1.7	-
randomised unpaired, fixed sample (1600 invocations)	0.05 (Bonferroni)	58.5 $\pm$ 1.7	-
live pilot chooser, raw / corrected ratio (measured on the Section 7 deployment; correction derived there)	-	1.39 / 1.22	-

Table 7. Paired against unpaired at k = 5, main effects, 6400 invocations, by task-difficulty spread. The mirror pays when tasks differ; it is unnecessary when they do not. The enumerated estimand is the same in every row to three decimals (0.0312, 0.0312, 0.0311), because the spreads share the same mean failure mass: the movement across rows is variance, not estimand.

task difficulty	mirror e-process, recovery	unpaired e-process, recovery	mirror FWER	unpaired FWER
homogeneous	87.1 $\pm$ 1.1	93.8 $\pm$ 0.9	0.8 $\pm$ 0.4	1.2 $\pm$ 0.6
Beta, concentration 4	91.8 $\pm$ 1.0	92.6 $\pm$ 0.9	0.2 ($\leq$1.2)	1.0 $\pm$ 0.5
Beta, concentration 1	96.9 $\pm$ 0.6	92.5 $\pm$ 0.9	0.0 ($\leq$0.7)	1.5 $\pm$ 0.6

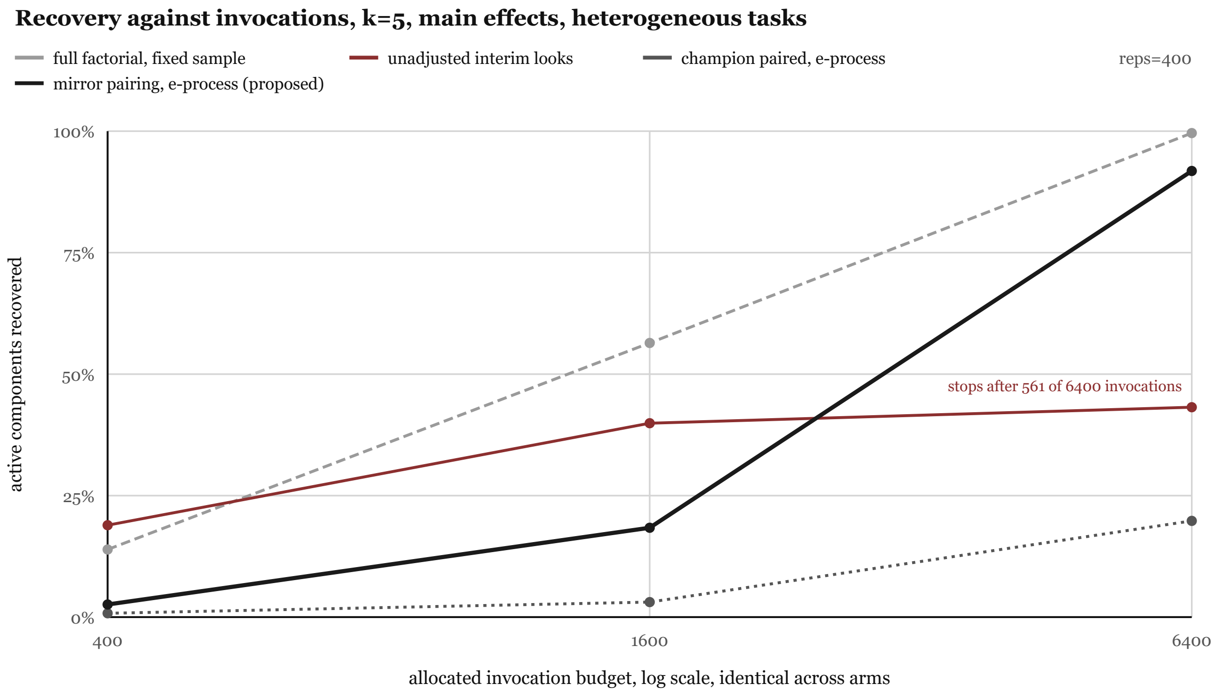

Figure 5. Recovery against invocations at k = 5. The interim arm consumes far fewer invocations because it stops early, which is exactly why its error rate is inflated.

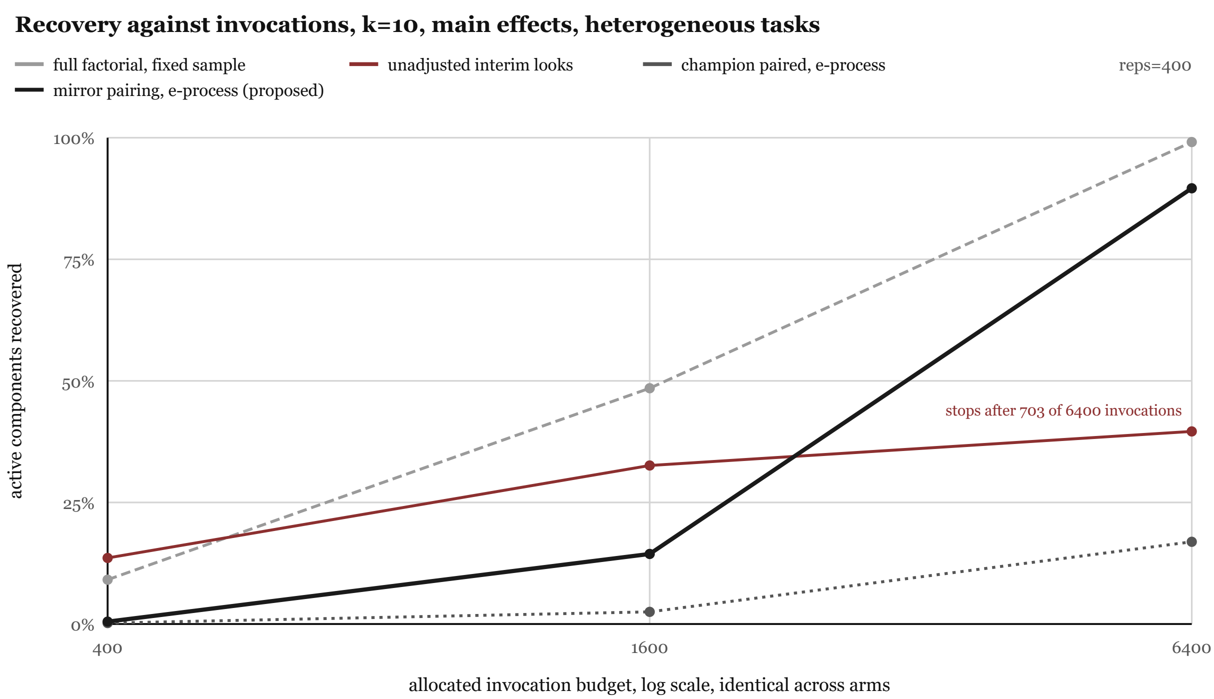

Figure 6. The same at k = 10, where enumerating the full factorial would require 1024 configurations. The cost of mirror screening does not grow with the number of configurations, only the threshold k/alpha does.

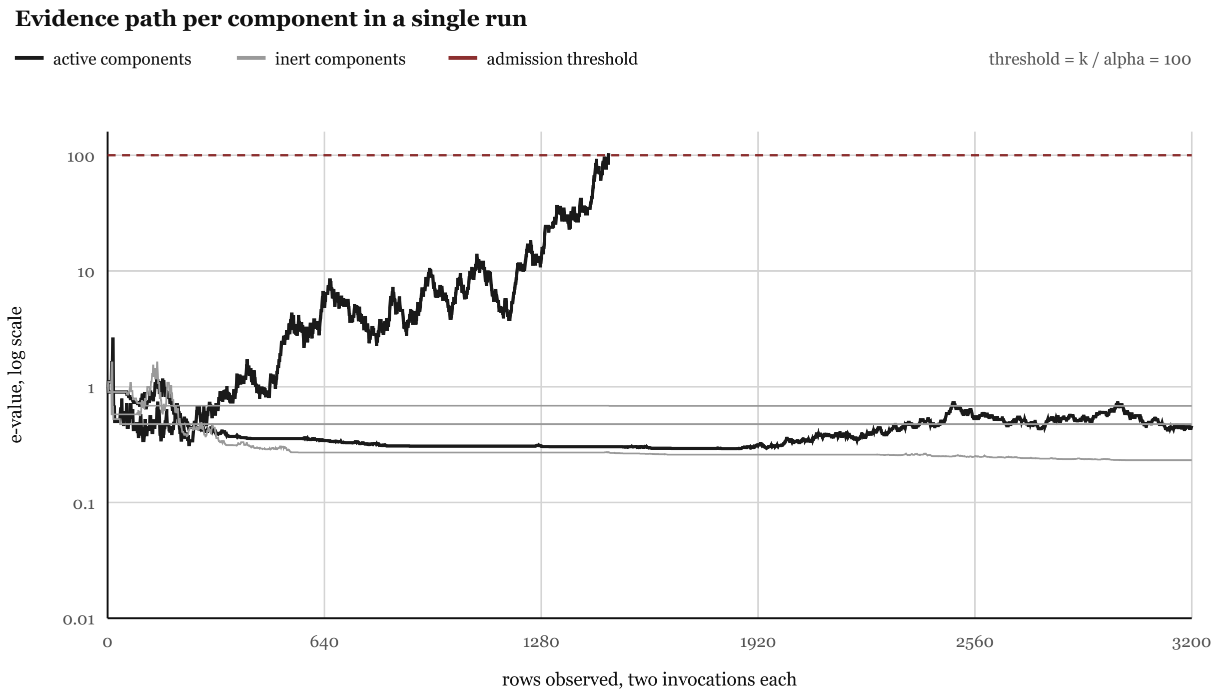

Figure 7. Evidence paths in a single run. One active component crosses the admission threshold and the run stops for that component; the second, with a smaller effect, has not accumulated enough evidence and remains undecided. Inert components drift downward.

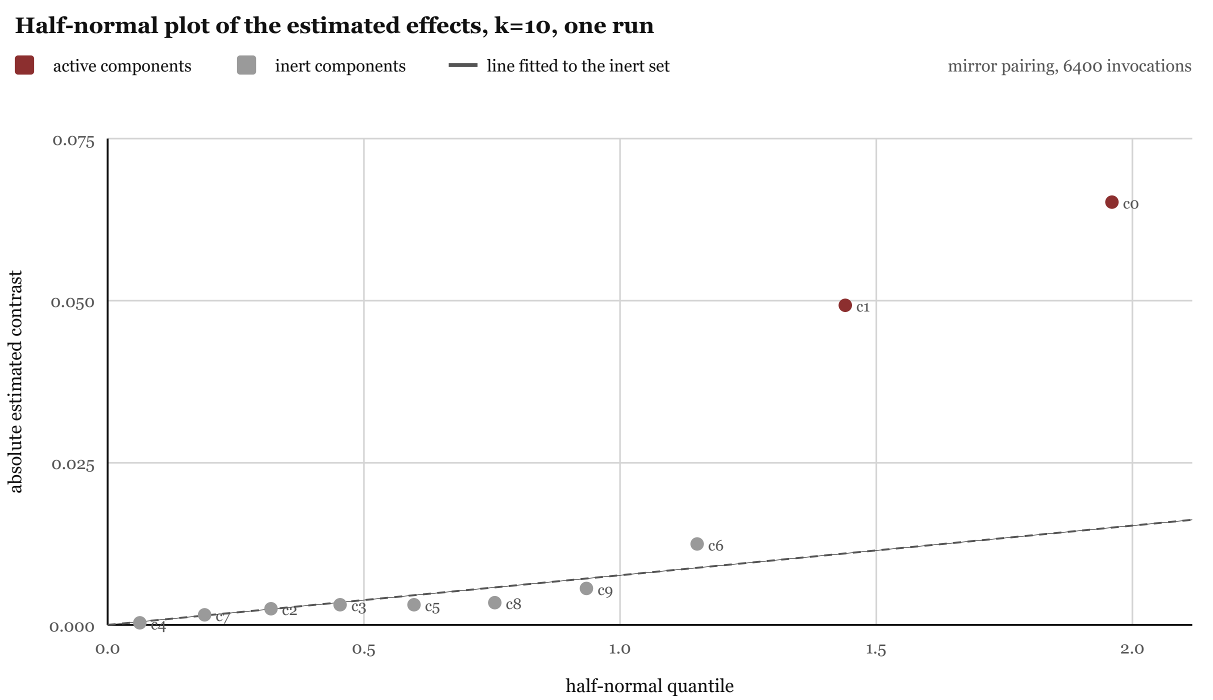

Figure 8. The standard screening plot for one run at ten components: absolute estimated contrasts against half-normal quantiles. The eight inert components lie on a line through the origin, the two active ones leave it. This is the diagnostic a practitioner reads before deciding what to keep.

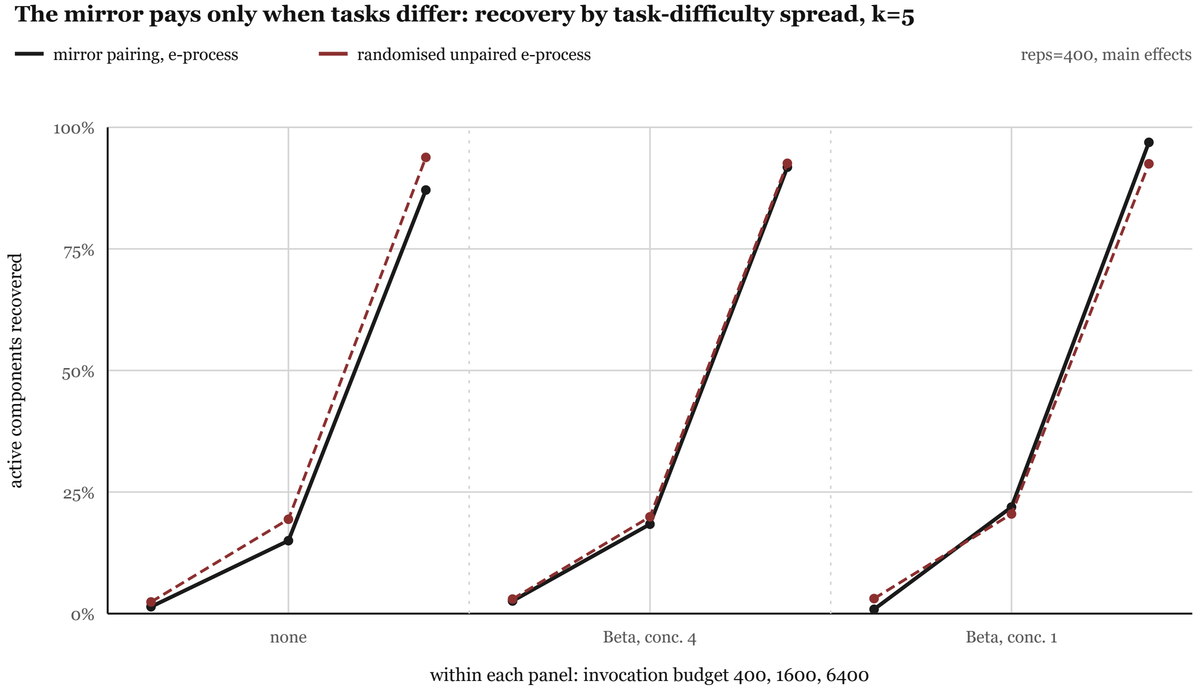

Figure 9. Recovery of the two active components by task-difficulty spread, k = 5, main effects. With homogeneous tasks the unpaired procedure matches the mirror at half the invocations per row; as the spread grows the task effect stays inside every unpaired increment and the mirror pulls away.

Table 8. The chooser under generative-model misspecification. Prediction is computed under the assumed model; the world is misspecified as stated. Admission rates are for the active component at equal invocation budget.

true world	**spread**	**chooser ratio (predicts)**	**admit rate mirror / unpaired**	**realized best**
heavy-tailed difficulty (20% extreme-mass mixture)	conc. 4	0.97 (unpaired)	75% / 67%	mirror
difficulty-correlated effects (easy tasks gain more)	conc. 4	0.97 (unpaired)	45% / 44%	mirror (within noise)
additive response (no multiplicative failure mass)	conc. 4	0.97 (unpaired)	100% / 100%	tie (within noise)
heavy-tailed difficulty (20% extreme-mass mixture)	conc. 1	1.15 (mirror)	85% / 72%	mirror
difficulty-correlated effects (easy tasks gain more)	conc. 1	1.15 (mirror)	12% / 10%	mirror (within noise)
additive response (no multiplicative failure mass)	conc. 1	1.15 (mirror)	100% / 100%	mirror (within noise)

Table 10. Graded verifier (score in [0,1], measurement noise sigma = 0.05), k = 5, budgets as marked per row, 400 replications, standard errors at the replication level. The estimand is the enumerated contrast of the binary tables, 0.0312 per active component.

task difficulty	**mirror, recovery**	**unpaired, recovery**	**mirror FWER (null)**	**unpaired FWER (null)**
homogeneous (400 invocations)	0.0 ($\leq$0.7)	10.5 $\pm$ 1.1%	0.0 ($\leq$0.7)	0.0 ($\leq$0.7)
homogeneous (800 invocations)	7.4 $\pm$ 1.0%	99.0 $\pm$ 0.4%	0.0 ($\leq$0.7)	0.0 ($\leq$0.7)
Beta, conc. 4 (1600 invocations)	99.6 ($\geq$98.4)	96.9 $\pm$ 0.6%	0.0 ($\leq$0.7)	0.0 ($\leq$0.7)
Beta, conc. 1 (1600 invocations)	98.6 $\pm$ 0.5%	73.6 $\pm$ 1.6%	0.0 ($\leq$0.7)	1.0 $\pm$ 0.5%

Table 11. Proposition 5 read against the two cells built to probe it: 6400 invocations, 400 replications, heterogeneous tasks, k = 5 shown.

cell	**tracked component vs the null**	**sequential screen admits**	**fixed-design marginal procedure admits**
sign change (outside the null)	helps before commitment, hurts after	10.8%	97.2%
all-stage non-positive (null true)	helps in no reachable context	0.0%	0.0%

Table 12. Every conditional-null-true cell behind the abstract's worst-case admission claim: the full cond-null grid over component count, task-difficulty spread and budget, 400 replications each, admission at the mirror e-process arm. No cell is summarised away; the promised level is 5%.

components	**spread**	**budget**	**admission %**
k10	hetero	400	0.0%
k10	hetero	1,600	0.0%
k10	hetero	6,400	0.0%
k10	homog	400	0.0%
k10	homog	1,600	0.0%
k10	homog	6,400	0.0%
k10	xhetero	400	0.0%
k10	xhetero	1,600	0.0%
k10	xhetero	6,400	0.0%
k5	hetero	400	0.2%
k5	hetero	1,600	0.0%
k5	hetero	6,400	0.0%
k5	homog	400	0.0%
k5	homog	1,600	0.0%
k5	homog	6,400	0.0%
k5	xhetero	400	0.0%
k5	xhetero	1,600	0.0%
k5	xhetero	6,400	0.0%

Table 14. The flip-confinement bound checked in the flip world: δ = 0.15, flip at row 60, 4,000 replications (exp_flip_bound.json). Every admission the flip actually buys happens at or before it, which is what Table 11 measures.

quantity	**value**
admissions at or before the flip	58
paths still active at the flip	3,942
realised post-flip admission rate	0.43%
averaged conditional bound min{1, E(τ)α/k}	2.79%

Table 15. Division of labour in the composed loop, with the measured evidence for each role. The three existing systems keep their jobs; the judge is the missing stage.

stage	**system**	**role kept**	**measured basis**
propose	SkillOpt-family protocol (published representative: reflective prompt evolution, Agrawal et al., 2025)	bounded edits, protected regions, rejected-candidate memory	its acceptance rule fails at a computable rate: exact round-level false adoption 0.965 to 0.999 (Proposition 10)
execute	agent runtime (DeepSeek Harness in our live run)	tools, sessions, recovery; same-task paired replay	450 live mirrored pairs (900 invocations) completed with ledger resume, Section 7
context	NOOA-style preview handles	hands large objects to the proposer as handles	screened as a component rather than assumed; the digest switch in Section 7
judge	this procedure	admission and pruning with family-wise control at any stopping time	0.5 to 1.8% false adoption at equal budget (Table 16)

Table 16. Acceptance rules for proposed edits, 6400 invocations, 400 replications. False adoption = at least one edit with no effect adopted in the round; null = all candidates inert; live = two of the candidates carry a genuine effect of 0.15.

acceptance rule	**false adoption, k=5 null**	**false adoption, k=10 null**	**false adoption, k=10 live**	**recovery, k=10 live**
point comparison (adopt if better)	96.8 ± 0.9%	100.0 (≥99.3)	99.2 ± 0.4%	95.4 ± 0.7%
point comparison, 2pp margin	75.8 ± 2.1%	98.2 ± 0.7%	95.8 ± 1.0%	88.4 ± 1.2%
per-candidate z-test, no correction	21.8 ± 2.1%	37.5 ± 2.4%	35.0 ± 2.4%	54.6 ± 1.8%
mirror screen (this paper)	1.8 ± 0.7%	0.8 ± 0.4%	0.5 (≤1.6)	88.1 ± 1.1%

Table 17. Live verdicts at 450 pairs (admission threshold k/alpha = 100). The effect column is the outcome-scale effect in percentage points with a variance-adaptive anytime confidence sequence (empirical-Bernstein, time-uniform, per-switch delta = 0.05) for the running average of the realised conditional means, which is a descriptive interval and not a guarantee about the final committed context; the last column reports closed testing on the pruning family's e-values (step-down, strong family-wise control, uniformly at least as powerful as the k/alpha threshold).

switch	verdict	rows	effect pp [95% anytime CS]	fixed-sample 95% CI	evidence for	evidence against	close
digest	undecided (running)	450	+5.1 [-18.4, +28.6]	[+0.6, +9.6]	3.77	1.00	
search_hint	undecided (running)	450	+3.8 [-19.7, +27.3]	[-0.8, +8.3]	1.49	0.80	
reflect	undecided (running)	450	-5.1 [-28.6, +18.4]	[-9.6, -0.6]	1.00	3.80	
fewshot	undecided (running)	450	+0.2 [-23.3, +23.8]	[-4.3, +4.8]	0.71	0.72	
strict	pruned	199	-14.6 [-59.7, +30.6]	[-21.7, -7.4]	1.00	115.91	

Table 22. Objections we consider strongest, and where each is addressed.

objection	status	response
The mirror estimator is not new	conceded	Antithetic variates, simultaneous perturbation and mirrored sampling all use it. The contribution is the conditional null a committing screen can promise, the per-component sequential admission that delivers it, and the measurement of how the standard alternatives fail.
Synthetic evidence for the method comparison	conceded in part	The comparison of procedures is simulation with measured parameters and published estimands; the procedure itself also ran live (Section 7) and produced one error-controlled pruning on a production agent.
Mirroring can produce a configuration nobody would deploy	conceded in part	Components may be pinned; only the flipped subset needs to be mirrored, at the cost of screening a subset at a time.
The estimand averages over the other components	conceded	Stated where it is derived. Pin the rest to obtain the deployed-configuration effect.
Two invocations per row is wasteful	conceded on binary verifiers	The unpaired one-invocation e-process with predictable centring is valid and is the default recommendation; the mirror is chosen only when the per-invocation log-growth ratio of Section 6 exceeds one, which on binary verifiers requires strong task-difficulty spread and on graded verifiers is typical.
The bet size has free parameters	answered	Nine settings of the cap and the bet stabiliser c (distinct from the futility floor f = 0.05 of Algorithm 1) were run at both component counts. Under the global null the family-wise error stayed at or below 1.5% in every one of the eighteen cells, mean 0.4%. Recovery moved with the stabiliser, from 75.0% to 93.2% at ten components, so the knob buys power and cannot buy validity. The headline cells use the pre-registered default c = 0.1, kept for guard margin; the best-recovery ablation setting (c = 0.05) is reported here, not silently adopted.
A union bound over components is conservative	answered	Closed testing over intersection e-value averages (step-down) is implemented and reported in Table 17; it is uniformly at least as powerful as the union-bound threshold and retains strong family-wise control. An e-BH variant for false discovery rate is the remaining extension.
Admit or prune is not a shipping decision without an effect size	answered	Table 17 reports, per component, the outcome-scale effect with a variance-adaptive anytime confidence sequence (empirical-Bernstein, time-uniform), readable at any stopping time including after the verdict.
The motivating citations are recent preprints	conceded	The harness-effect literature is months old; the build resolves identifiers but cannot confer peer review on them. The claims we rely on quantitatively are our own measurements.

Table 23. What a practitioner takes from this paper, as rules.

situation	**rule**	**measured basis**
choosing which harness switches stay	randomise levels per task, one e-process per switch, stop whenever you like	FWER at nominal in every null cell (admission and pruning each at their own alpha; acting on both reads 2 alpha); interim looks without this inflate error to 29-42% at 6400 invocations
tempted by a fractional design	do not: randomisation gives the same cheapness without the alias	88-94% wrong-component admission under interaction
accepting proposed prompt edits	never on a point comparison, whatever the margin	0.96-0.999 round-level false adoption, computed exactly
pair with mirror or run unpaired	estimate between-task spread from a pilot; pair only when the log-growth ratio exceeds one - on graded verifiers it usually does, but not at every budget (Table 10 prints the 800-invocation exception)	six of seven paired-versus-unpaired cells resolved as predicted, one within noise (Tables 8 and 11)
reading verdicts after early commitments	the guarantee is the conditional null; a component that helps in no committed context is admitted at most alpha; a component whose effect may flip after a commitment carries no guarantee and should be re-screened with the others pinned at the committed levels	worst 0.2% across 18 cells with the null true